On supplementary variable methods

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1 Using Supplementary Variable Method

Traditionally we have been solving only the system of governing equations while making sure that the discretized numerical scheme respects energy dissipation. But here we employ a technique wherein we introduce auxiliary equations that are already consistent with the original equations. Typically these equations are definitions of the free energy and the time-invariance of its dissipation rate. Solving these overall set of equations in a systematic manner results in a solution that well respects the energy structure of the physical system. However, in thlivis process we end up making the system over-determined. Thus we add certain perturbation variables called supplementary variables to make the system well-determined. This method is called Supplementary Variable Method (SVM).

1.1 Illustrating the SVM method

In [1], Gong et al. discuss the SVM method in detail for a gradient flow model. In it, the mobility equation is perturbed with an additive term, which potentially results in a structurally stable method.

$$\begin{cases} \phi_t = -M\left(\mathcal{L}\phi + \frac{\delta f}{\delta \phi}\right) + \alpha g[\phi], \\ \frac{d}{dt}F[\phi] = -(\mu, M\mu) \\ F[\phi] = \frac{1}{2}(\phi, \mathcal{L}\phi) + (f(\phi, \nabla\phi), 1) \\ \mu = \mathcal{L}\phi + \frac{\delta f}{\delta \phi} \end{cases} \tag{1.1}$$

A predictor-corrector scheme is designed on this system:

Scheme 1.1. Given Φ^{n-1} and Φ^n, we compute Φ^{n+1} through the following two steps:

1. Prediction:

$$\frac{\Phi_*^{n+\frac{1}{2}} - \Phi^n}{\tau/2} = -\mathbf{M}\left(\mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\overline{\Phi}^{n+\frac{1}{2}}\right]\right) \tag{1.2}$$

2. SVM correction:

$$\delta_t^+ \Phi^n = -\mathbf{M}\left(\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) + \alpha g[\Phi_*^{n+\frac{1}{2}}], \tag{1.3}$$

$$\delta_t^+ F\left[\Phi^n\right] = -\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right), \tag{1.4}$$

$$F[\Phi] = \frac{1}{2}(\Phi, \mathcal{L}\Phi) + (f(\Phi, \nabla\Phi), 1), \tag{1.5}$$

$$\mu_*^{n+\frac{1}{2}} = \mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right] \tag{1.6}$$

1.2 How to "meaningfully" add the supplementary variable

There are endless choices to add the supplementary variable to a given PDE system. Here we seek a systematic and insightful way to perturb the system.

First, recall that for any PDE that has been written down, we need to analyze certain intrinsic aspects of it, viz.,

- PDE-level well-posedness
  - existence and uniqueness of solutions
  - Continuous dependence of the solution on its ICs and BCs
  - energy structure preservation (energy stability)
- for its numerical schemes:
  - stability
  - convergence (mesh refinement)
  - consistency (conformity with the exact solution)
- symmetry and invariance
- variational formulations, etc.

1.3 Using the error terms in the numerical scheme

One potential way is to analyze the error term in the numerical scheme. The idea is to analyze residual error in the numerical scheme which could give us a hint on what the supplementary variable can look like. Let's look at some examples:

1.3.1 1d Heat equation - perturbing the mobility equation

Consider the 1d heat equation with suitable I.C and B.Cs.

$$u_t = \kappa u_{xx}$$

Coupling this with the equations for free energy and its dissipation, we get the following system:

$$\begin{cases} u_t = \kappa u_{xx}, \\ \mathcal{E}(t) = \frac{1}{2}\int_\Omega u^2 dx, \\ \frac{d}{dt}\mathcal{E}[u] = -\int_\Omega u_x^2 dx \end{cases} \tag{1.7}$$

Discretize the system

$$\begin{cases} \dfrac{u_i^{n+1} - u_i^n}{\Delta t} = \kappa \dfrac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{(\Delta x)^2} & \text{(1.8a)} \\ \mathcal{E}^n = \dfrac{1}{2}\sum (u_i^n)^2 \Delta x & \text{(1.8b)} \\ \dfrac{\mathcal{E}^{n+1} - \mathcal{E}^n}{\Delta t} = -\sum \left(\dfrac{u_{i+1}^n - u_{i-1}^n}{2\Delta x}\right)^2 & \text{(1.8c)} \end{cases}$$

Using Taylor expansion on (1.8a):

- LHS

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\partial u}{\partial t}(x_i, t^n) + \frac{\Delta t}{2!}\frac{\partial^2 u}{\partial t^2}(x_i, \xi^n)$$

- RHS

$$\begin{aligned} \tfrac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{(\Delta x)^2} &= \tfrac{2}{2!}\tfrac{\partial^2 u}{\partial x^2}(x_i, t^{n+1}) + \tfrac{2(\Delta x)^2}{4!}\tfrac{\partial^4 u}{\partial x^4}(\chi_i, t^{n+1}) \\ &= \left(\tfrac{\partial^2 u}{\partial x^2}(x_i, t^n) + \tfrac{\Delta t}{1!}\tfrac{\partial^3 u}{\partial t \partial x^2}(x_i, \theta^n)\right) + \tfrac{2(\Delta x)^2}{4!}\tfrac{\partial^4 u}{\partial x^4}(\chi_i, t^{n+1}) \\ &= \left(\tfrac{\partial^2 u}{\partial x^2}(x_i, t^n) + \kappa\tfrac{\Delta t}{1!}\tfrac{\partial^4 u}{\partial x^4}(x_i, \theta^n)\right) + \tfrac{(\Delta x)^2}{2}\tfrac{\partial^4 u}{\partial x^4}(\chi_i, t^{n+1}) \end{aligned}$$

, where $\chi_i \in [x_i, x_{i+1}]$ and $\xi^n, \theta^n \in [t_i, t_{i+1}]$.

Substituting these expansions back into the scheme (1.8a), we get:

$$\left(\frac{\partial u}{\partial t}(x_i, t^n) - \kappa\frac{\partial^2 u}{\partial x^2}(x_i, t^n)\right) = \kappa^2 \Delta t \frac{\partial^4 u}{\partial x^4}(x_i, \theta^n) + \kappa\frac{(\Delta x)^2}{2}\frac{\partial^4 u}{\partial x^4}(\chi_i, t^{n+1}) - \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2}(x_i, \xi^n)$$

So the error term in the RHS above gives us hints about how we can perturb the heat equation when using BTCS scheme.

The heat equation can be perturbed as such:

$$u_t = \kappa u_{xx} + \alpha(u_{xxxx} + u_{tt})$$

Thus the system that must be solved is:

$$\begin{cases} u_t = \kappa u_{xx} + \alpha(u_{xxxx} + u_{tt}), \\ \mathcal{E}(t) = \frac{1}{2}\int_\Omega u^2 dx, \\ \frac{d}{dt}\mathcal{E}[u] = -\int_\Omega u_x^2 dx \end{cases} \tag{1.9}$$

Here we have 3 unknowns $\{u, \mathcal{E}, \alpha\}$ and 3 equations, which makes it a consistent system.

Linear stability analysis:

It must be noted that adding supplementary variables must be done such that it doesn't disturb the stability of the system. Here we analyze the stability behavior of adding the 2 terms into the 1d heat equation.

$$u_t = \kappa u_{xx} + \theta_1 u_{xxxx} + \theta_2 u_{tt}$$

Take the Fourier transform

$$\theta_2 \frac{d^2U}{dt^2} - \frac{dU}{dt} + (\theta_1 k^4 - \theta_2 \kappa k^2) = 0$$

$$\theta_2 r^2 - r + (\theta_1 k^4 - \theta_2 \kappa k^2) = 0$$

For the system to be stable, we want its roots to have negative real part so that if $e^{rt} \to 0$ as $t \to \infty$. Its discriminant D $= -4\theta_1\theta_2 k^4 + 4\kappa\theta_2 k^2 + 1$

- **Complex roots case**: We strictly need $D < 0$. So the quadratic D(k) must have negative leading coeff and non-real roots. $\implies \theta_1\theta_2 > 0$ and $\theta_2(\theta_2\kappa^2 + \theta_1) < 0$

  Additionally, for stability, the r-quadratic must have roots with negative real parts. $\implies \frac{1}{\theta_2} < 0 \implies \theta_2 < 0$, which leads to a contradiction $\Rightarrow\Leftarrow$

- **both negative real roots** $\implies r_1 + r_2 < 0, r_1 r_2 > 0$ and $D \geq 0$

$$\theta_2 < 0, \frac{(\theta_1 k^4 - \theta_2 \kappa k^2)}{\theta_2} > 0 \text{ and } -4\theta_1\theta_2 k^4 + 4\kappa\theta_2 k^2 + 1 \geq 0$$

$$\theta_2 < 0, (\theta_1 k^4 - \theta_2 \kappa k^2) < 0 \text{ and } \theta_1\theta_2 < 0, \theta_2(\theta_2\kappa^2 + \theta_1) \leq 0$$

  This leads to a contradiction $\Rightarrow\Leftarrow$

- **both positive real roots case must be avoided**:

  $\implies r_1 + r_2 > 0, r_1 r_2 > 0$ and $D \geq 0$

$$\theta_2 > 0, \frac{(\theta_1 k^4 - \theta_2 \kappa k^2)}{\theta_2} > 0 \text{ and } -4\theta_1\theta_2 k^4 + 4\kappa\theta_2 k^2 + 1 \geq 0$$

$$\theta_2 > 0, (\theta_1 k^4 - \theta_2 \kappa k^2) > 0 \text{ and } \theta_1\theta_2 < 0, \theta_2(\theta_2\kappa^2 + \theta_1) \leq 0$$

  This leads to a contradiction $\Rightarrow\Leftarrow$. So this case will never happen

- **real roots with opposite signs**:

  $\implies r_1 r_2 < 0$ and $D \geq 0$

$$\frac{(\theta_1 k^4 - \theta_2 \kappa k^2)}{\theta_2} < 0 \text{ and } -4\theta_1\theta_2 k^4 + 4\kappa\theta_2 k^2 + 1 \geq 0$$

$$\frac{(\theta_1 k^4 - \theta_2 \kappa k^2)}{\theta_2} < 0 \text{ and } \theta_1\theta_2 < 0, \theta_2(\theta_2\kappa^2 + \theta_1) \leq 0$$

This is possible but only when $\theta_1\theta_2 < 0, \theta_2(\theta_2\kappa^2 + \theta_1) \leq 0$. Additionally, for stability we want the positive root's term must vanish to 0. Which means the initial conditions must be chosen carefully.

1.3.2 Using SVM on Allen-Cahn equation - perturbing the chemical potential

Consider the general energy dissipative gradient flow system:

$$\begin{cases} \phi_t = -M \cdot \mu, \\ \mu = \frac{\delta F}{\delta \phi} = \mathcal{L}\phi + \frac{\delta f}{\delta \phi} \\ F[\phi] = \frac{1}{2}(\phi, \mathcal{L}\phi) + (f(\phi, \nabla\phi), 1) \\ \frac{d}{dt}F[\phi] = -(\mu, M\mu) \end{cases} \tag{1.10}$$

This system could be perturbed as such:

$$\begin{cases} \phi_t = -M\left(\mathcal{L}\phi + \frac{\delta f}{\delta \phi}\right) + \alpha g_1[\phi], \\ \mu = \frac{\delta F}{\delta \phi} = \mathcal{L}\phi + \frac{\delta f}{\delta \phi} + \alpha g_2[\phi] \\ F[\phi] = \frac{1}{2}(\phi, \mathcal{L}\phi) + (f(\phi, \nabla\phi), 1) + \alpha g_3[\phi] \\ \frac{d}{dt}F[\phi] = -(\mu, M\mu) \end{cases} \tag{1.11}$$

Let us consider a specific case of the above system: the Allen-Cahn system. $f(\phi) = \frac{1}{4}(1 - \phi^2)^2, \mathcal{L}(\cdot) = -\epsilon^2\Delta(\cdot), M = I$

$$\begin{cases} \phi_t = \epsilon^2\Delta\phi - (\phi^3 - \phi) + \alpha g_1[\phi], \\ \mu = -\epsilon^2\Delta\phi + (\phi^3 - \phi) + \alpha g_2[\phi] \\ F[\phi] = \frac{1}{2}(\phi, -\epsilon^2\Delta\phi) + (\phi^3 - \phi, 1) + \alpha g_3[\phi] \\ \frac{d}{dt}F[\phi] = -(\mu, \mu) \end{cases} \tag{1.12}$$

Consider the discretized scheme of the mobility equation in 1d space:

$$\frac{\phi_i^{n+1} - \phi_i^n}{\Delta t} = -\epsilon^2\left(\frac{\phi_{i+1}^{n+1} - 2\phi_i^{n+1} + \phi_{i-1}^{n+1}}{(\Delta x)^2}\right) + \left((\phi^{n+1})^3 - (\phi^{n+1})\right)$$

Using Taylor expansion and computing the residual term in the numerical equation:

$$\frac{\partial \phi^n}{\partial t} - \epsilon^2\phi_{xx}^n + ((\phi^n)^3 - (\phi^n)) = \left[-\frac{\Delta t}{2}\phi_{tt}^n + \epsilon^2\phi_{txx}^n - 3(\phi^n)^2\phi_t^n\Delta t - 3\phi^n(\phi_t^n)^2(\Delta t)^2 - (\phi_t^n)^3(\Delta t)^3 + \phi_t^n\Delta t\right]$$

Adding these residual error terms to the PDE needs further analysis. Doing Fourier stabiliy analysis on it is not straight forward. All in all, we can say that even though adding the residual error terms of the effective PDE seems natural, we must resolve these 2 challenges:

- The added terms must not induce any unstable modes into the system. And performing this analyzing is not straightforward in cases with nonlinear chemical potential
- the added terms must be carefully handled during formulating the discrete numerical scheme

1.4 Perturbing the Mobility operator

Let's consider other ways that may potentially insinuate how to add the perturbation terms. Here we look at what happens when we perturb the mobility operator.

Let $\mathbf{N}$ be a positive semi-definite operator like $\mathbf{M}$. Here we perturb the mobility operator $\mathbf{M} \to \mathbf{M} + \alpha^2 \mathbf{N}$

$$\begin{cases} \phi_t = -(\mathbf{M} + \alpha^2 \mathbf{N}) \cdot \mu = -\mathbf{M} \cdot \mu - \alpha^2 \mathbf{N} \cdot \mu, \\ \mu = \frac{\delta F}{\delta \phi} = \mathcal{L}\phi + \frac{\delta f}{\delta \phi} \\ F[\phi] = \frac{1}{2}(\phi, \mathcal{L}\phi) + (f(\phi, \nabla\phi), 1) \\ \frac{d}{dt} F[\phi] = -(\mu, (\mathbf{M} + \alpha^2 \mathbf{N})\mu) = -(\mu, \mathbf{M}\mu) - \alpha^2 (\mu, \mathbf{N}\mu) \end{cases} \tag{1.13}$$

With these discrete operator definitions, considering the below predictor-corrector scheme:

$$\overline{\Phi}^{n+\frac{1}{2}} := \frac{3\Phi^n - \Phi^{n-1}}{2}, \qquad \Phi^{n+\frac{1}{2}} := \frac{\Phi^{n+1} + \Phi^n}{2}$$

Here we introduce the following predictor-corrector numerical scheme:

Scheme 1.2. Given Φ^{n-1} and Φ^n, we compute Φ^{n+1} through the following two steps:

1. Prediction: predict $\Phi_*^{n+\frac{1}{2}}$ via

$$\frac{\Phi_*^{n+\frac{1}{2}} - \Phi^n}{\tau/2} = -\mathbf{M}\left(\mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\overline{\Phi}^{n+\frac{1}{2}}\right]\right) \tag{1.14}$$

2. SVM correction:

$$\begin{cases} \delta_t^+ \Phi^n = -\mathbf{M}\left(\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) - \alpha^2 \mathbf{N}\left(\mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right), \\ \delta_t^+ F\left[\Phi^n\right] = -\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right) - \alpha^2\left(\mu_*^{n+\frac{1}{2}}, \mathbf{N}\mu_*^{n+\frac{1}{2}}\right), \\ F[\Phi] = \frac{1}{2}(\Phi, \mathcal{L}\Phi) + (f(\Phi, \nabla\Phi), 1), \\ \mu_*^{n+\frac{1}{2}} = \mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right] \end{cases} \tag{1.15}$$

This Scheme (1.2) preserves the discrete energy dissipation law by design.

$$F\left[\Phi^{n+1}\right]-F\left[\Phi^{n}\right]=-\tau\left(\mu_{*}^{n+\frac{1}{2}},(\mathbf{M}+\alpha^{2}\mathbf{N})\mu_{*}^{n+\frac{1}{2}}\right)\leq 0 \tag{1.16}$$

Doing local truncation error analysis on (1.14), we have

$$\Phi_{*}^{n+\frac{1}{2}}=\Phi(t_{n+\frac{1}{2}})+\mathcal{O}(\tau^{2}) \tag{1.17}$$

Now outline the strategy to solve the Scheme (1.2). Eq. (1.14) can be rewritten as

$$\Phi_{*}^{n+\frac{1}{2}}=\left(1+\frac{\tau}{2}\mathbf{M}\mathcal{L}\right)^{-1}\left(\Phi^{n}-\frac{\tau}{2}\mathbf{M}\frac{\delta f}{\delta\Phi}[\overline{\Phi}^{n+\frac{1}{2}}]\right) \tag{1.18}$$

After getting the predicted value $\Phi_{*}^{n+\frac{1}{2}}$, we move to the correction part.
The first equation of (1.15) can be rewritten as

$$\Phi^{n+1}=\widehat{\Phi}^{n+1}+\beta w^{n}, \tag{1.19}$$

where

$$\widehat{\Phi}^{n+1}=\left(1+\frac{\tau}{2}\mathbf{M}\mathcal{L}\right)^{-1}\left(\left(1-\frac{\tau}{2}\mathbf{M}\mathcal{L}\right)\Phi^{n}-\tau\mathbf{M}\frac{\delta f}{\delta\Phi}[\Phi_{*}^{n+\frac{1}{2}}]\right) \tag{1.20a}$$

$$w^{n}=-\left(1+\frac{\tau}{2}\mathbf{M}\mathcal{L}\right)^{-1}\left[\mathcal{L}\Phi_{*}^{n+\frac{1}{2}}+\frac{\delta f}{\delta\Phi}[\Phi_{*}^{n+\frac{1}{2}}]\right] \tag{1.20b}$$

$$\beta=-\tau\alpha^{2}. \tag{1.20c}$$

Then we plug Φ^{n+1} into the second equation in (1.15) to obtain

$$F\left[\widehat{\Phi}^{n+1}+\beta w^{n}\right]=F\left[\Phi^{n}\right]-\tau\left(\mu_{*}^{n+\frac{1}{2}},\mathbf{M}\mu_{*}^{n+\frac{1}{2}}\right)-\beta\left(\mu_{*}^{n+\frac{1}{2}},\mathbf{N}\mu_{*}^{n+\frac{1}{2}}\right), \tag{1.21}$$

which is an algebraic equation for β. In general, it can have multiple solutions, but one of them must be close to 0 and it approaches to zero as $\tau\to 0$. So, we solve for this solution using an iterative method such as the Newton iteration with 0 as the initial condition, it generally converges to a solution close to 0 when τ is not too large. After obtaining β, we can evaluate Φ^{n+1}. The existence of solution β is discussed in the following theorem.

Theorem 1.1. *If* $-\left(\frac{\delta F}{\delta\Phi}[\Phi^{n}],\mathbf{N}(\mathcal{L}\Phi^{n}+\frac{\delta f}{\delta\phi}\Phi^{n})\right)+\left(\mu_{*}^{n+\frac{1}{2}},\mathbf{N}\mu_{*}^{n+\frac{1}{2}}\right)\neq 0$, *there exists a* $\tau^{*}>0$ *such that (1.16) defines a unique function* $\beta=\beta(\tau)$ *for all* $\tau\in[0,\tau^{*}]$ *and Scheme 1.2 is second order.*

Proof. Proof. For τ,β in a neighborhood of $(0,0)$, we define the real function

$$u(\tau,\beta)=F[\widehat{\Phi}^{n+1}+\beta w^{n}]-F[\Phi^{n}]+\tau\left(\mu_{*}^{n+\frac{1}{2}},\mathbf{M}\mu_{*}^{n+\frac{1}{2}}\right)+\beta\left(\mu_{*}^{n+\frac{1}{2}},\mathbf{N}\mu_{*}^{n+\frac{1}{2}}\right)$$

It follows from Eqns (1.18 - 1.20b) that

$$u(0,0)=0, \quad \frac{\partial u}{\partial \beta}(0,0)=-\left(\frac{\delta F}{\delta \Phi}[\Phi^n], \mathbf{N}\left[\mathcal{L}\Phi_*^{n+\frac{1}{2}}+\frac{\delta f}{\delta \phi}\Phi_*^{n+\frac{1}{2}}\right]\right)+\left(\mu_*^{n+\frac{1}{2}}, \mathbf{N}\mu_*^{n+\frac{1}{2}}\right)\neq 0$$

According to the implicit function theorem, there exists a $\tau^* > 0$ such that the equation $u(\tau,\beta)=0$ defines a unique smooth function $\beta=\beta(\tau)$ satisfying $\beta(0)=0$ and $u(\tau,\beta(\tau))=0$ for all $\tau\in[-\tau^*,\tau^*]$. □

Since $\widehat{\Phi}^{n+1}$ satisfies the following scheme

$$\frac{\widehat{\Phi}^{n+1}-\Phi^n}{\tau}=-\mathbf{M}\left(\mathcal{L}\left(\frac{\widehat{\Phi}^{n+1}+\Phi^n}{2}\right)+\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right)$$

through local truncation error analysis we have

$$\widehat{\Phi}^{n+1}=\Phi\left(t_{n+1}\right)+\mathcal{O}\left(\tau^3\right), \quad F\left[\widehat{\Phi}^{n+1}\right]=F\left[\Phi\left(t_{n+1}\right)\right]+\mathcal{O}\left(\tau^3\right)$$

where (1.17) was used. In addition, we expand

$$u(\tau,\beta)=u(\tau,0)+\beta\frac{\partial u}{\partial \beta}(\tau,0)+\mathcal{O}\left(\beta^2\right)$$

with

$$u(\tau,0)=F\left[\widehat{\Phi}^{n+1}\right]-F\left[\Phi^n\right]+\tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right)=\mathcal{O}\left(\tau^3\right)$$

$$\frac{\partial u}{\partial \beta}(\tau,0)=\frac{\partial u}{\partial \beta}(0,0)+\mathcal{O}(\tau)=\left(\frac{\delta F}{\delta \Phi}\left[\Phi^n\right], g\left[\Phi^n\right]\right)+\mathcal{O}(\tau)$$

Then $\beta=\beta(\tau)=\mathcal{O}\left(\tau^3\right)$ and the proposed scheme is of order 2

1.5 Perturbing the Friction operator

In the above method, we have considered a perturbation in the mobility M. Consider a more generic framework which includes the friction operator too.

Consider the kinetic equation

$$\Phi_t=-D\mu$$

The operator D can be factorized into $D=W\cdot M$, where W is invertible under appropriate conditions. So the above kinetic equation now becomes:

$$C\Phi_t=-M\mu, \quad C=W^{-1}$$

Hence the 'general' gradient flow system now becomes:

$$\begin{cases} C\phi_t=-M\cdot\mu, \\ \mu=\frac{\delta F}{\delta \phi}=\mathcal{L}\phi+\frac{\delta f}{\delta \phi} \\ F[\phi]=\frac{1}{2}(\phi,\mathcal{L}\phi)+(f(\phi,\nabla\phi),1) \\ \frac{d}{dt}F[\phi]=-(\mu,M\mu) \end{cases} \tag{1.22}$$

Similar to the above methods, now let's perturb the friction operator $C \to C - \alpha^2 I$, with I being the identity. In the following proof, we consider a 1D space. We get:

$$C\phi_t - \alpha^2 \phi_t = -M \cdot \mu.$$

Considering this predictor-corrector scheme:

Scheme 1.3 (Semi-implicit perturbation)**.** Given Φ^{n-1} and Φ^n, we compute Φ^{n+1} through the following two steps:

1. Prediction: predict $\Phi_*^{n+\frac{1}{2}}$ via

$$C\left(\frac{\Phi_*^{n+\frac{1}{2}} - \Phi^n}{\tau/2}\right) = -\mathbf{M}\left(\mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\overline{\Phi}^{n+\frac{1}{2}}\right]\right) \tag{1.23}$$

2. SVM correction:

$$\begin{cases} C(\delta_t^+ \Phi^n) - \alpha^2\left(\dfrac{\Phi_*^{n+\frac{1}{2}} - \Phi^n}{\tau/2}\right) = -\mathbf{M}\left(\mathcal{L}\Phi^{n+\frac{1}{2}} + \dfrac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right), \\ \delta_t^+ F\left[\Phi^n\right] = -\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right), \\ F[\Phi] = \dfrac{1}{2}(\Phi, \mathcal{L}\Phi) + (f(\Phi, \nabla\Phi), 1), \\ \mu_*^{n+\frac{1}{2}} = \mathcal{L}\Phi_*^{n+\frac{1}{2}} + \dfrac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right] \end{cases} \tag{1.24}$$

Note that we consider the perturbation term $-\alpha^2 \Phi_t$ explicitly.
This Scheme (**??**) preserves the discrete energy dissipation law by design.

$$F\left[\Phi^{n+1}\right] - F\left[\Phi^n\right] = -\tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right) \leq 0 \tag{1.25}$$

Now let's continue discussing the solvabilty strategy for the Scheme (**??**):
Doing local truncation error analysis on (1.23), we have

$$\Phi_*^{n+\frac{1}{2}} = \Phi(t_{n+\frac{1}{2}}) + \mathcal{O}(\tau^2) \tag{1.26}$$

Here we outline the strategy to solve the Scheme (**??**). Eq. (1.23) can be rewritten as

$$\Phi_*^{n+\frac{1}{2}} = \left(C + \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)^{-1}\left(C\Phi^n - \frac{\tau}{2}\mathbf{M}\frac{\delta f}{\delta \Phi}[\overline{\Phi}^{n+\frac{1}{2}}]\right) \tag{1.27}$$

After getting the predicted value $\Phi_*^{n+\frac{1}{2}}$, we move to the correction part.

The first equation of (1.24) can be rewritten as

$$\Phi^{n+1} = \widehat{\Phi}^{n+1} + \beta w^n, \tag{1.28}$$

where

$$\widehat{\Phi}^{n+1} = \left(C + \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)^{-1} \left(\left(C - \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)\Phi^n - \tau\mathbf{M}\frac{\delta f}{\delta\Phi}[\Phi_*^{n+\frac{1}{2}}]\right) \tag{1.29a}$$

$$w^n = -\left(C + \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)^{-1}\Phi^n \tag{1.29b}$$

$$\beta = 2\alpha^2. \tag{1.29c}$$

Then we plug Φ^{n+1} into the second equation in (1.24) to obtain

$$F\left[\widehat{\Phi}^{n+1} + \beta w^n\right] = F\left[\Phi^n\right] - \tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right), \tag{1.30}$$

which is an algebraic equation for β. In general, it can have multiple solutions, but one of them must be close to 0 and it approaches to zero as $\tau \to 0$. So, we solve for this solution using an iterative method such as the Newton iteration with 0 as the initial condition, it generally converges to a solution close to 0 when τ is not too large. After obtaining β, we can evaluate Φ^{n+1}. The existence of solution β is discussed in the following theorem.

Theorem 1.2. *If $\left(\frac{\delta F}{\delta\Phi}[\Phi^n], C^{-1}\Phi^n\right) \neq 0$, there exists a $\tau^* > 0$ such that (1.30) defines a unique function $\beta = \beta(\tau)$ for all $\tau \in [0, \tau^*]$ and Scheme 1.3 is second order.*

Proof. Proof. For τ, β in a neighborhood of $(0, 0)$, we define the real function

$$u(\tau, \beta) = F[\widehat{\Phi}^{n+1} + \beta w^n] - F[\Phi^n] + \tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right)$$

It follows from Eqns (1.27 - 1.29b) that

$$u(0,0) = 0, \quad \frac{\partial u}{\partial\beta}(0,0) = -\left(\frac{\delta F}{\delta\Phi}[\Phi^n], C^{-1}\Phi^n\right) \neq 0$$

According to the implicit function theorem, there exists a $\tau^* > 0$ such that the equation $u(\tau, \beta) = 0$ defines a unique smooth function $\beta = \beta(\tau)$ satisfying $\beta(0) = 0$ and $u(\tau, \beta(\tau)) = 0$ for all $\tau \in [-\tau^*, \tau^*]$. □

Since $\widehat{\Phi}^{n+1}$ satisfies the following scheme

$$\frac{\widehat{\Phi}^{n+1} - \Phi^n}{\tau} = -\mathbf{M}\left(\mathcal{L}\frac{\widehat{\Phi}^{n+1} + \Phi^n}{2} + \frac{\delta f}{\delta\Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right)$$

through local truncation error analysis we have

$$\widehat{\Phi}^{n+1} = \Phi\left(t_{n+1}\right) + \mathcal{O}\left(\tau^3\right), \quad F\left[\widehat{\Phi}^{n+1}\right] = F\left[\Phi\left(t_{n+1}\right)\right] + \mathcal{O}\left(\tau^3\right)$$

where (1.26) was used. In addition, we expand

$$u(\tau,\beta) = u(\tau,0) + \beta\frac{\partial u}{\partial \beta}(\tau,0) + \mathcal{O}\left(\beta^2\right)$$

with

$$u(\tau,0) = F\left[\widehat{\Phi}^{n+1}\right] - F\left[\Phi^n\right] + \tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right) = \mathcal{O}\left(\tau^3\right)$$
$$\frac{\partial u}{\partial \beta}(\tau,0) = \frac{\partial u}{\partial \beta}(0,0) + \mathcal{O}(\tau) = \left(\frac{\delta F}{\delta \Phi}\left[\Phi^n\right], g\left[\Phi^n\right]\right) + \mathcal{O}(\tau)$$

Then $\beta = \beta(\tau) = \mathcal{O}\left(\tau^3\right)$ and the proposed scheme is of order 2

1.5.1 Alternate way: treating the α^2 perturbation term implicity

In the first eqn in the system (1.24), we treat the perturbation term explicitly. If instead we treated it implicitly:

$$(C-\alpha^2)(\delta_t^+\Phi^n) = -\mathbf{M}\left(\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right)$$

So the overall scheme becomes:

Scheme 1.4 (Implicit perturbation term)**.** Given Φ^{n-1} and Φ^n, we compute Φ^{n+1} through the following two steps:

1. Prediction: predict $\Phi_*^{n+\frac{1}{2}}$ via

$$C\left(\frac{\Phi_*^{n+\frac{1}{2}} - \Phi^n}{\tau/2}\right) = -\mathbf{M}\left(\mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\overline{\Phi}^{n+\frac{1}{2}}\right]\right) \tag{1.31}$$

2. SVM correction:

$$\begin{cases} (C-\alpha^2)(\delta_t^+\Phi^n) = -\mathbf{M}\left(\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right), \\ \delta_t^+ F\left[\Phi^n\right] = -\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right), \\ F[\Phi] = \frac{1}{2}(\Phi, \mathcal{L}\Phi) + (f(\Phi,\nabla\Phi),1), \\ \mu_*^{n+\frac{1}{2}} = \mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right] \end{cases} \tag{1.32}$$

Strategy to solve this system :

From Eq. (1.31), we can get $\Phi_*^{n+\frac{1}{2}}$

$$\Phi_*^{n+\frac{1}{2}} = \left(C + \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)^{-1}\left[\Phi^n - \frac{\tau}{2}\mathbf{M}\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right]$$

Next, in order to solve for Φ^{n+1}: Simplifying the first equation in (1.32), we get

$$\Phi^{n+1} = (C - \alpha^2 I + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\left((C - \alpha^2 I - \frac{\tau}{2}M\mathcal{L})\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) \tag{1.33}$$

If we let:

$$\widehat{\Phi}^{n+1} = (C - \alpha^2 I + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\left((C - \frac{\tau}{2}M\mathcal{L})\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) \tag{1.34a}$$

$$w^n = (C - \alpha^2 I + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\Phi^n \tag{1.34b}$$

$$\beta = -\alpha^2 \tag{1.34c}$$

So we have

$$\Phi^{n+1} = \widehat{\Phi}^{n+1} + \beta w^n \tag{1.35}$$

Substituting this in the third equation in (1.31) yields:

$$F\left[\widehat{\Phi}^{n+1} + \beta w^n\right] = F\left[\Phi^n\right] - \tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right). \tag{1.36}$$

This equation can be solved using a nonlinear solver. Since we want $\beta \sim 0$, in the non-linear solver we start with an initial seed of 0 and expect that the solution converges. However this demands a rigorous proof to ensure existence of the a solution β that is close to 0. Once we obtain β we use eq. (1.35) to get Φ^{n+1}.

Theorem 1.3. *If $\left(\frac{\delta F}{\delta \Phi}[\Phi^n], C^{-1}\Phi^n\right) \neq 0$, there exists a $\tau^* > 0$ such that (1.36) defines a unique function $\beta = \beta(\tau)$ for all $\tau \in [0, \tau^*]$ and Scheme 1.4 is second order.*

Proof. Proof. For τ, β in a neighborhood of $(0, 0)$, we define the real function

$$u(\tau, \beta) = F[\widehat{\Phi}^{n+1} + \beta w^n] - F[\Phi^n] + \tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right)$$

$$\left.\frac{\partial u}{\partial \beta}\right|_{(0,0)} = \left.\left(\frac{\partial F}{\partial \beta}\left[\widehat{\Phi}^{n+1} + \beta w^n\right], \frac{\partial}{\partial \beta}(\widehat{\Phi}^{n+1} + \beta w^n)\right)\right|_{(0,0)}$$

Note that:

$$\left.\frac{\partial F}{\partial \beta}\left[\widehat{\Phi}^{n+1} + \beta w^n\right]\right|_{(0,0)} = \frac{\partial F}{\partial \beta}\left[\Phi^n\right] \tag{1.37}$$

and

$$\left.\frac{\partial}{\partial \beta}(\widehat{\Phi}^{n+1} + \beta w^n)\right|_{(0,0)} = \left[-[C + \beta I + \frac{\tau}{2}M\mathcal{L}]^{-2}\left((C - \frac{\tau}{2}M\mathcal{L})\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right)\right. \tag{1.38}$$

$$+ \beta[C + \beta I + \frac{\tau}{2} M\mathcal{L}]^{-1}\Phi^n\Big]\Big|_{(0,0)} \tag{1.39}$$

$$= -C^{-2}C\Phi^n \tag{1.40}$$

$$= -C^{-1}\Phi^n. \tag{1.41}$$

$$\implies \frac{\partial u}{\partial \beta}\bigg|_{(0,0)} = \left(\frac{\delta F}{\delta \Phi}[\Phi^n], C^{-1}\Phi^n\right)$$

Thus,

$$u(0,0) = 0, \quad \frac{\partial u}{\partial \beta}(0,0) = -\left(\frac{\delta F}{\delta \Phi}[\Phi^n], C^{-1}\Phi^n\right) \neq 0$$

According to the implicit function theorem, there exists a $\tau^* > 0$ such that the equation $u(\tau, \beta) = 0$ defines a unique smooth function $\beta = \beta(\tau)$ satisfying $\beta(0) = 0$ and $u(\tau, \beta(\tau)) = 0$ for all $\tau \in [-\tau^*, \tau^*]$. Since the time step is non-negative, we can limit this permissible bound to $[0, \tau^*]$. □

Comment: The existence of a β that is close to 0 is proved by the above theorem. However the efficiency of solving is debatable. In the previous versions of perturbations, the definitions of $\widehat{\Phi}$ and w^n did not contain the model variable β or α. So the could be obtained by solving linear equations individually. In this case, the equations (1.34a-1.34b) can't be solved individually. The only way of obtaining β is by solving (1.36), which when expanded out gives this:

$$\begin{aligned} 0 = F\Bigg[&(C + \beta I + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\left((C - \frac{\tau}{2}M\mathcal{L})\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) \\ + \quad &\beta(C + \beta I + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\Phi^n\Bigg] \\ - F\left[\Phi^n\right] + \tau&\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right) \end{aligned}$$

1.5.2 Alternate way of dealing with the inverse operator

Recall the equation to be solved (1.33)

$$\Phi^{n+1} = (C + \beta I + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\left((C + \beta I - \frac{\tau}{2}M\mathcal{L})\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right)$$

Here, since the inverse operator is not devoid of the variable to be solved (β), there is no way of breaking down this into neatly solvable linear systems of equations. As a result, when solving the non-linear equation for β, it is leading to inaccurate solutions, possibly due to the complex gradient evaluation. So we do a workaround here where we use a Power series expansion of the inverse operator and approximate it.

Let:

- $A = (C + \frac{\tau}{2}M\mathcal{L})$

- $B = (C - \frac{\tau}{2} M\mathcal{L})$

$$\Phi^{n+1} = (A + \beta I)^{-1} \left((B + \beta I)\Phi^n - \tau M \frac{\delta f}{\delta \Phi} \left[\Phi_*^{n+\frac{1}{2}} \right] \right)$$

Expanding the inverse and collecting the coefficients of powers of β together:

$$\begin{aligned}
\Phi^{n+1} = & \left(A^{-1} B \Phi^n - \tau A^{-1} M \frac{\delta f}{\delta \Phi} \left[\Phi_*^{n+\frac{1}{2}} \right] \right) \\
& + \beta \left((A^{-1} - (A^{-1})^2 B) \Phi^n \right) \\
& + \beta^2 \left(-(A^{-1})^2 + (A^{-1})^3 B) \Phi^n - \tau (A^{-1})^3 M \frac{\delta f}{\delta \Phi} \left[\Phi_*^{n+\frac{1}{2}} \right] \right)
\end{aligned}$$

Substituting this in the third equation in (1.31) yields:

$$F\left[\Phi^{n+1}\right] = F\left[\Phi^n\right] - \tau \left(\mu_*^{n+\frac{1}{2}}, \mathbf{M} \mu_*^{n+\frac{1}{2}} \right). \tag{1.42}$$

This equation can be solved using a nonlinear solver. Compared to the previous way of decomposing Φ^{n+1}, here we only have powers of β with known coefficients.This will ease up the computation when computing the gradient for the Newton's method. Once we obtain β we use eq. (1.35) to get Φ^{n+1}.

Theorem 1.4. *If* $\left(\frac{\delta F}{\delta \Phi}[\Phi^n], A^{-1}\Phi^n\right) \neq 0$*, there exists a* $\tau^* > 0$ *such that the energy equation admits a unique solution* $\beta = \beta(\tau)$ *for all* $\tau \in [0, \tau^*]$*.*

Proof. Proof. Consider the asymptotic expansion of Φ^{n+1} in powers of β:

$$\Phi^{n+1} = \Phi_0 + \beta \Phi_1 + \beta^2 \Phi_2 + O(\beta^3)$$

where Φ_0, Φ_1, and Φ_2 are defined above.

For τ, β in a neighborhood of $(0, 0)$, we define the real function

$$u(\tau, \beta) = F[\Phi_0 + \beta \Phi_1 + \beta^2 \Phi_2] - F[\Phi^n] + \tau \left(\mu_*^{n+\frac{1}{2}}, \mathbf{M} \mu_*^{n+\frac{1}{2}} \right)$$

Taking the partial derivative with respect to β:

$$\left. \frac{\partial u}{\partial \beta} \right|_{(0,0)} = \left. \left(\frac{\partial F}{\partial \beta} \left[\Phi_0 + \beta \Phi_1 + \beta^2 \Phi_2 \right], \Phi_1 + 2\beta \Phi_2 \right) \right|_{(0,0)}$$

Note that:

$$\left. \frac{\partial F}{\partial \beta} \left[\Phi_0 + \beta \Phi_1 + \beta^2 \Phi_2 \right] \right|_{(0,0)} = \frac{\partial F}{\partial \beta} \left[A^{-1} B \Phi^n - \tau A^{-1} \mathbf{M} \frac{\delta f}{\delta \Phi} \right] \tag{1.43}$$

and

$$\Phi_1\Big|_{(0,0)} = (A^{-1} - (A^{-1})^2 B)\Phi^n \tag{1.44}$$

$$= -A^{-1}\Phi^n \tag{1.45}$$

Therefore:

$$\implies \frac{\partial u}{\partial \beta}\Big|_{(0,0)} = \left(\frac{\delta F}{\delta \Phi}[\Phi^n], A^{-1}\Phi^n\right)$$

Thus,

$$u(0,0) = 0, \quad \frac{\partial u}{\partial \beta}(0,0) = -\left(\frac{\delta F}{\delta \Phi}[\Phi^n], A^{-1}\Phi^n\right) \neq 0$$

By the implicit function theorem, there exists a $\tau^* > 0$ such that the equation $u(\tau, \beta) = 0$ defines a unique smooth function $\beta = \beta(\tau)$ satisfying $\beta(0) = 0$ and $u(\tau, \beta(\tau)) = 0$ for all $\tau \in [0, \tau^*]$. The uniqueness of this solution ensures the well-posedness of the scheme. □

The problem with this is that the evaluation of Φ^{n+1} deals with explicit inversion of the banded matrix A. So we seek alternate ways of perturbing the system.

1.6 Alternate way of perturbing

1.6.1 Scaling the friction matrix

Start with the mobility equation and perturbing the friction operator:

$$C(1-\alpha^2)\delta_t^+\Phi^n = -M\left[\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right]$$

This leads to:

$$\left(C(1-\alpha^2) + \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)\Phi^{n+1} = \left((C(1-\alpha^2) - \frac{\tau}{2}\mathbf{M}\mathcal{L})\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right)$$

The variable α still appears in the LHS operator. So won't work.

1.6.2 Perturbing multiple terms

Let us start with the $\Phi^{n+1}-$ equation and work backward. We desire this following equation as it can be broken down into multiple solvable linear systems like in the original paper:

$$(1-\alpha^2)\left(C + \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)\Phi^{n+1} = \left((1-\alpha^2)\left(C - \frac{\tau}{2}\mathbf{M}\mathcal{L}\right)\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) \tag{1.46}$$

This leads to:

$$C(1-\alpha^2)\delta_t^+\Phi^n = -M\left[(1-\alpha^2)\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right]$$

So essentially, we are perturbing (scaling) the friction operator C and the operator $\mathcal{L}$.
Eq. (1.46) can be broken down into this:

$$\widehat{\Phi}^{n+1} = (C + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\left((C - \frac{\tau}{2}M\mathcal{L})\Phi^n - \tau M\frac{\delta f}{\delta\Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) \tag{1.47a}$$

$$w^n = (C + \frac{\tau}{2}\mathbf{M}\mathcal{L})^{-1}\left((C - \frac{\tau}{2}M\mathcal{L})\Phi^n\right) \tag{1.47b}$$

$$\beta = -\alpha^2 \tag{1.47c}$$

So we have

$$\Phi^{n+1} = \frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n \tag{1.48}$$

Eqns. (1.47a) and (1.47b) are two linear systems which can be solved using a suitable solver. Finally, β could be solved using (1.48) and the dissipation equation:

$$F\left[\Phi^{n+1}\right] = F\left[\Phi^n\right] - \tau\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right). \tag{1.49}$$

Theorem 1.5. *If* $\left(\frac{\delta F}{\delta\Phi}[\Phi^n], w^n - \widehat{\Phi}^{n+1}\right) \neq 0$, *there exists a* $\tau > 0$ *such that equation (1.49) defines a unique function* $\beta = \beta(\tau)$ *for all* $\tau \in [0, \tau]$.

Proof. For τ, β in a neighborhood of $(0,0)$, we define the real function:

$$u(\tau,\beta) = F\left[\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n\right] - F[\Phi^n] + \tau\left(\mu^{n+\frac{1}{2}}, \mathbf{M}\mu^{n+\frac{1}{2}}\right)$$

$$\left.\frac{\partial u}{\partial\beta}\right|_{(0,0)} = \left(\frac{\delta F}{\delta\Phi}\left[\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n\right], \frac{\partial}{\partial\beta}\left(\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n\right)\right)\Bigg|_{(0,0)}$$

When $\tau = 0$, we have $\widehat{\Phi}^{n+1} = \Phi^n$ and $w^n = \Phi^n$ from equations (1.47a) and (1.47b), which implies that for any β, $\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n = \Phi^n$. Therefore, $u(0,0) = 0$.

Now we compute the partial derivative of u with respect to β at $(0,0)$:

$$\left.\frac{\partial u}{\partial\beta}\right|(0,0) = \left(\frac{\delta F}{\delta\Phi}\left[\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n\right], \frac{\partial}{\partial\beta}\left(\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n\right)\right)\Bigg|(0,0)$$

For the derivative of the second term:

$$\begin{aligned}\frac{\partial}{\partial\beta}\left(\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1} + \frac{\beta}{(1+\beta)}w^n\right) &= \frac{\partial}{\partial\beta}\left(\frac{1}{(1+\beta)}\widehat{\Phi}^{n+1}\right) + \frac{\partial}{\partial\beta}\left(\frac{\beta}{(1+\beta)}w^n\right) \\ &= -\frac{1}{(1+\beta)^2}\widehat{\Phi}^{n+1} + \frac{(1+\beta)-\beta}{(1+\beta)^2}w^n\end{aligned}$$

$$= -\frac{1}{(1+\beta)^2}\widehat{\Phi}^{n+1} + \frac{1}{(1+\beta)^2}w^n$$
$$= \frac{1}{(1+\beta)^2}(w^n - \widehat{\Phi}^{n+1})$$

Evaluating at $(0,0)$, and noting that $\frac{\delta F}{\delta \Phi}[\Phi^n]$ is the variational derivative at $\tau = 0$:

$$\left.\frac{\partial u}{\partial \beta}\right|_{(0,0)} = \left.\left(\frac{\delta F}{\delta \Phi}[\Phi^n], w^n - \widehat{\Phi}^{n+1}\right)\right|_{(0,0)} = \left(\frac{\delta F}{\delta \Phi}[\Phi^n], \Phi^n - \Phi^n\right)$$
$$\frac{\partial u}{\partial \beta}(0,0) = 0$$

So the implicit function theorem may not be applied. It's possible that there are other intuitive ways of perturbing.

[Work in progress..] □

1.6.3 General way of perturbing

Let P, Q be scalar multiples, but they can be generalized to be differential operators too.

Consider this general way of perturbing:

$$PC\delta_t^+\Phi^n = -M\left(Q\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right) \tag{1.50}$$

Rearranging to get the Φ^{n+1} equation:

$$A\Phi^{n+1} = B\Phi^n - \tau M\frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right], \tag{1.51}$$

where

$$A = PC + Q\left(\frac{\tau M\mathcal{L}}{2}\right), \qquad B = PC - Q\left(\frac{\tau M\mathcal{L}}{2}\right)$$
$$\implies P = \left(\frac{A+B}{2}\right)C^{-1}, \quad Q = (M\mathcal{L})^{-1}\left(\frac{A-B}{\tau}\right)$$

The idea is to start with (1.51), choose suitable {A, B} that will aid us in efficiently solving for Φ^{n+1}. This will guide us on how we can perturb the dynamic equation (1.50).

For example: if we choose:

$$A = (1-\alpha^2)\left(C + \frac{\tau M\mathcal{L}}{2}\right), \qquad B = (1-\alpha^2)\left(C - \frac{\tau M\mathcal{L}}{2}\right),$$
$$\text{this yields } P = \left(1-\alpha^2\right), \quad Q = (1-\alpha^2).$$

this will give us the following perturbed equation:

$$C(1-\alpha^2)\delta_t^+\Phi^n = -M\left((1-\alpha^2)\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right),$$

which is the method that was discussed in in the above section (1.6.3).

So, now with this idea, we need to cleverly choose A, B to get a viable method that will work for us.

2 Numerical results

We first solve the 2d Allen Cahn system using the scheme described in scheme (1.2).

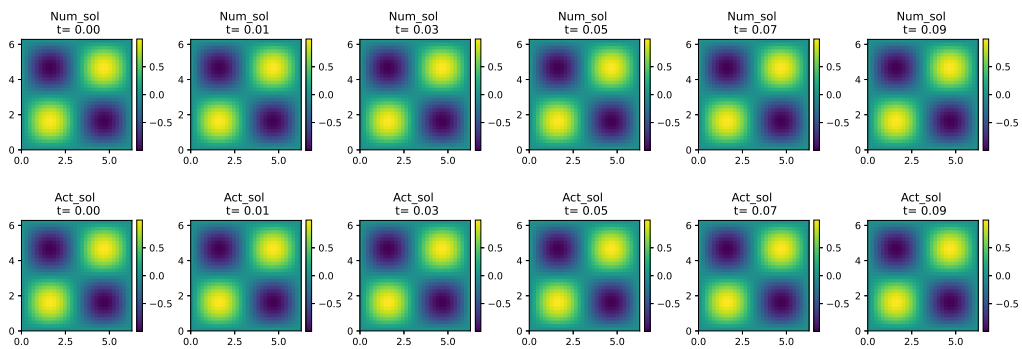

Figure 2.1: Numerical vs actual solution using Scheme (1.4)

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2.1 A generic approach of perturbing the friction operator

[this is the older framework where the perturbation terms are additive to the friction operator C]

Thus far, we have seen above how perturbing the friction operator works in 1D. Let's generalize the perturbation further.

Consider the gradient flow with a generalized perturbation term χ:

$$(C - \chi)\partial_t \Phi = -M\mu, \qquad \chi = \sum_{i=0}^{n} \alpha_i \nabla^{2i}$$

We discuss 2 types of perturbations:

- Type-1 stabilizer (n=0)
$$(C - \alpha_0 I)\partial_t \Phi = -M\mu$$
- Type-2 stabilizer (n=1)
$$(C - \alpha_0 I - \alpha_1 \nabla^2)\partial_t \Phi = -M\mu$$

Now the question one may ask is: how does the free energy of the system get modified by adding the above perturbation terms? So let us follow a systematic way of arriving at the free energy functional.

First let us revisit and see how the energy if derived in an unperturbed gradient flow

2.1.1 Energy dynamics in a general gradient flow:

Suppose we start with the mobility equation

$$\partial_t \Phi = -M(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi})$$

Let's try to derive the free energy from this. Multiply with $(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi})$ on both the sides and integrate $\int_\Omega$.

$$\int_\Omega (\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi})\partial_t \Phi = -\int_\Omega (\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi}) M(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi})$$

So

$$\int_\Omega \frac{\delta}{\delta \Phi}\left(\frac{1}{2}\Phi\mathcal{L}\Phi + f[\Phi]\right)\frac{\partial \Phi}{\partial t} = -\left(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi}, M(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi})\right)$$

$$\frac{d}{dt}\left(\int_\Omega \frac{1}{2}\Phi\mathcal{L}\Phi + f[\Phi]\right) = -\left(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi}, M(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi})\right)$$

Thus we can term the Free energy of the gradient flow system as:

$$F[\Phi] = \int_\Omega \left(\frac{1}{2}\Phi\mathcal{L}\Phi + f[\Phi]\right)$$

And its dissipation rate is given by:

$$\frac{dF}{dt} = -\left(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi}, M(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi})\right)$$

This leads to the complete consistent system:

$$\begin{cases} C\Phi_t = -M \cdot \mu \\ F[\Phi] = \frac{1}{2}(\Phi, \mathcal{L}\Phi) + (f, 1) \\ \mu = \frac{\delta F}{\delta \Phi} = \mathcal{L}\Phi + \frac{\delta f}{\delta \Phi} \\ \frac{d}{dt}F[\Phi] = -(\mu, M\mu) \end{cases} \tag{2.1}$$

Perturbing the friction operator:

Upon perturbing the friction operator in the mobility equation, we get:

$$C\Phi_t = -M(\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi}) \longrightarrow (C - \sum_{i=0}^{n} \alpha_i \nabla^{2i})\tilde{\Phi}_t = -M(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})$$

This perturbation of the mobility operator results in a whole different system with modified state variables.

$$[\Phi, \mu, F] \longrightarrow [\tilde{\Phi}, \tilde{\mu}, \tilde{F}]$$

The objective of applying SVM here is to be able to have a consistent perturbed system which can be solved numerically in an 'optimal' manner. Then letting $\alpha_i = 0, \forall i$, we can recover the solution of the original unperturbed system.

Now given that the mobility equation is perturbed as above, we need to figure out the energy functional of the new system.

2.1.2 Energy dynamics in type-1 stabilizer:

Let's look at the energy dynamics in Type-1 stabilizer here:

$$(C - \alpha_0 I)\partial_t \tilde{\Phi} = - - M(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})$$

In order to have a dissipative model, we need $\alpha_0 < 0$. So let's say $\alpha_0 = -\alpha^2$. In the standard gradient flows, we have the friction coeff $C = I$. This yields:

$$\partial_t \tilde{\Phi} = -M\left(\frac{1}{1+\alpha^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right)$$

Multiplying both sides by $\left(\frac{1}{1+\alpha^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right)$ and integrating over Ω

$$\frac{d}{dt}\left(\frac{1}{1+\alpha^2}\int_\Omega \frac{1}{2}\tilde{\Phi}\mathcal{L}\tilde{\Phi} + f[\tilde{\Phi}]\right) = -\left(\left(\frac{1}{1+\alpha^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right), M\left(\frac{1}{1+\alpha^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right)\right) \leq 0$$

Thus we can term the free energy of the perturbed gradient flow system as:

$$\tilde{F}[\tilde{\Phi}] = \left(\frac{1}{1+\alpha^2}\int_\Omega \frac{1}{2}\tilde{\Phi}\mathcal{L}\tilde{\Phi} + f[\tilde{\Phi}]\right)$$

And its dissipation rate is given by:

$$\frac{d\tilde{F}}{dt} = -\left(\left(\frac{1}{1+\alpha^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right), M\left(\frac{1}{1+\alpha^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right)\right)$$

This leads to the complete system:

$$\begin{cases} (C - \alpha_0 I)\partial_t \tilde{\Phi} = -M\tilde{\mu} \\ \tilde{F}[\tilde{\Phi}] = \dfrac{1}{1+\alpha^2}\displaystyle\int_\Omega \left(\frac{1}{2}\tilde{\Phi}\mathcal{L}\tilde{\Phi} + f[\tilde{\Phi}]\right) \\ \tilde{\mu} = \dfrac{\delta \tilde{F}}{\delta \tilde{\Phi}} = \dfrac{1}{1+\alpha^2}\left(\mathcal{L}\tilde{\Phi} + \dfrac{\delta f}{\delta \tilde{\Phi}}\right) \\ \dfrac{d}{dt}\tilde{F}[\tilde{\Phi}] = -(\tilde{\mu}, M\tilde{\mu}) \end{cases} \tag{2.2}$$

2.1.3 Energy dynamics in type-2 stabilizer:

Let's look at the Type-2 stabilizer here:

$$(C - \alpha_0 I - \alpha_1 \frac{\partial}{\partial x^2})\partial_t \tilde{\Phi} = -M(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})$$

Proof. In the standard gradient flows, we have the friction coeff $C = I$.

Multiplying both sides by $\left(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}}\right)$ and integrating over Ω

$$\int_\Omega \frac{\delta}{\delta \tilde{\Phi}} \left((1-\alpha_0)(\frac{1}{2}\Phi\mathcal{L}\Phi + f)\right)\tilde{\Phi}_t - \alpha_1 \int_\Omega (\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\tilde{\Phi}_{txx} = -\left((\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}}), M(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right) \leq 0$$

$$\frac{d}{dt}\left((1-\alpha_0)\int_\Omega (\frac{1}{2}\Phi\mathcal{L}\Phi + f)\right) - \alpha_1 \int_\Omega \frac{d^2}{dx^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\tilde{\Phi}_t + B.T = -\left((\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}}), M(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right) \leq 0$$

Define $G[\tilde{\Phi}]$ such that

$$\frac{\delta G[\tilde{\Phi}]}{\delta \tilde{\Phi}} := \frac{d^2}{dx^2}(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})$$

And ensuring that the boundary terms vanish:

$$[(\mathcal{L}\tilde{\Phi} + f')\tilde{\Phi}_{tx} - \frac{d}{dx}(\mathcal{L}\tilde{\Phi} + f')\tilde{\Phi}_t]\Big|_{\partial\Omega} = 0$$

$$\frac{d}{dt}\left((1-\alpha_0)\int_\Omega \left(\frac{1}{2}\Phi\mathcal{L}\Phi + f[\Phi]\right) - \alpha_1 G[\tilde{\Phi}]\right) = -\left((\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}}), M(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right) \leq 0$$

Thus we can term the free energy of the perturbed gradient flow system as:

$$\tilde{F}[\tilde{\Phi}] = (1-\alpha_0)\int_\Omega \left(\frac{1}{2}\Phi\mathcal{L}\Phi + f[\Phi]\right) - \alpha_1 G[\tilde{\Phi}]$$

And its dissipation rate is given by:

$$\frac{d\tilde{F}}{dt} = -\left((\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}}), M(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}})\right)$$

This leads to the consistent system:

$$\begin{cases} (C - \alpha_0 I - \alpha_1 \frac{\partial}{\partial x^2})\partial_t \tilde{\Phi} = -M\tilde{\mu} \\ \tilde{F}[\tilde{\Phi}] = (1-\alpha_0)\int_\Omega \left(\frac{1}{2}\Phi\mathcal{L}\Phi + f[\Phi]\right) - \alpha_1 G[\tilde{\Phi}] \\ \tilde{\mu} = \frac{\delta \tilde{F}}{\delta \tilde{\Phi}} = \left(\mathcal{L}\tilde{\Phi} + \frac{\delta f}{\delta \tilde{\Phi}}\right) \\ \frac{d}{dt}\tilde{F}[\tilde{\Phi}] = -(\tilde{\mu}, M\tilde{\mu}) \end{cases} \tag{2.3}$$

Thus far we have shown how perturbing the mobility equation

□

2.2 Stability of the physical system

Take the type-2 stabilizer. And for simplicity, let's first consider the 1-d space.

$$(1-\alpha_0-\alpha_1\frac{\partial}{\partial x^2})\partial_t\Phi = -M(\mathcal{L}\Phi + f(\Phi))$$

Let's consider the dynamics of the PDE around its equilibrium solution Φ^* (i.e., $(\Phi^*)_t = 0$). So we let $\Phi = \Phi^* + \delta\Phi$. Assuming $\mathcal{L}$ is a linear operator, we get:

$$(1-\alpha_0)(\delta\Phi)_t - \alpha_1(\delta\Phi)_{txx} = -M\mathcal{L}(\delta\Phi) - M\delta\Phi\left(\frac{\delta f}{\delta\Phi}[\Phi^*]\right)$$

Taking the Fourier transform on both sides:

- $\mathcal{F}((\delta\Phi)_t) = \frac{d}{dt}\mathcal{F}[\delta\Phi]$
- $\mathcal{F}((\delta\Phi)_{xxt}) = (ik)^2\frac{d}{dt}\mathcal{F}[\delta\Phi]$

$$(1-\alpha_0)\frac{d\mathcal{F}[\delta\Phi]}{dt} + \alpha_1 k^2\frac{d\mathcal{F}[\delta\Phi]}{dt} = -M\mathcal{F}[\mathcal{L}(\delta\Phi)] - M\mathcal{F}[\delta\Phi]\left(\frac{\delta f}{\delta\Phi}[\Phi^*]\right)$$

$$\frac{d\mathcal{F}[\delta\Phi]}{dt} = -M\left(\frac{\mathcal{F}[\mathcal{L}(\delta\Phi)] + \mathcal{F}[\delta\Phi]\left(\frac{\delta f}{\delta\Phi}[\Phi^*]\right)}{1-\alpha_0+\alpha_1 k^2}\right)$$

Based on the choce of the operator $\mathcal{L}$, we get a specific ODE, whose analysis can be performed to get the stability criterion.

For example: If $\mathcal{L}$ is the Laplacian, i.e., $\mathcal{L} = \frac{\partial^2}{\partial x^2}(\cdot)$

$$\mathcal{F}[\mathcal{L}(\delta\Phi)] = -k^2\mathcal{F}[\delta\Phi]$$

$$\therefore \frac{d\mathcal{F}[\delta\Phi]}{dt} = M\left(\frac{-k^2 - f'(\Phi^*)}{1-\alpha_0+\alpha_1 k^2}\right)\mathcal{F}[\delta\Phi]$$

For stability, we want the growth factor above to be negative. This can be ensured if $\frac{\delta f}{\delta\Phi}\big|_{\Phi^*} > 0$ and $\alpha_0 < 1, \alpha_1 > 0$.

In general for $\chi = \sum_{i=0}^{n}\alpha_i\nabla^{2i}$, linear stability can be ensured if the polynomial in k, $(1-\sum_{i=0}^{n}\alpha_i(ik)^i) > 0$ along with having $f'(\Phi^*) > 0$.

To generalize the given 1D PDE analysis to multiple dimensions, we'll work with vector fields and employ the gradient and Laplacian operators in multi-dimensional space. We'll also introduce the Jacobian matrix when considering the nonlinearity in the function $f(\cdot)$.

Generalization to Multi-Dimensional Space

Let's consider the operator $\mathcal{L}(\cdot) = -\nabla^2(\cdot) + f(\cdot)$ in a d-dimensional space where ∇^2 is the Laplacian operator and ∇ is the gradient operator. The PDE is then:

$$(1-\alpha_0-\alpha_1\nabla^2)\frac{\partial\Phi}{\partial t} = -M(-\nabla^2\Phi + f(\Phi)).$$

Assume the equilibrium solution Φ^* satisfies:

$$\mathcal{L}(\Phi^*) = -\nabla^2\Phi^* + f(\Phi^*) = 0.$$

Let $\Phi = \Phi^* + \delta\Phi$, and linearize the PDE around Φ^*:

$$(1-\alpha_0)\frac{\partial(\delta\Phi)}{\partial t} - \alpha_1\nabla^2\frac{\partial(\delta\Phi)}{\partial t} = M\nabla^2(\delta\Phi) - MJ_f(\Phi^*)\delta\Phi,$$

where $J_f(\Phi^*)$ is the Jacobian matrix of $f(\Phi)$ evaluated at Φ^*.
Taking the Fourier transform of the linearized PDE leads to:
- $\mathcal{F}[\nabla^2\delta\Phi] = -|\mathbf{k}|^2\mathcal{F}[\delta\Phi]$, - $\mathcal{F}[\nabla^2\frac{\partial\delta\Phi}{\partial t}] = -|\mathbf{k}|^2\frac{\partial}{\partial t}\mathcal{F}[\delta\Phi]$,
where $\mathbf{k}$ is the wave vector in Fourier space.
Substituting these into the linearized PDE gives:

$$(1-\alpha_0)\frac{\partial\mathcal{F}[\delta\Phi]}{\partial t} + \alpha_1|\mathbf{k}|^2\frac{\partial\mathcal{F}[\delta\Phi]}{\partial t} = -M|\mathbf{k}|^2\mathcal{F}[\delta\Phi] - MJ_f(\Phi^*)\mathcal{F}[\delta\Phi].$$

Rearranging terms, we have:

$$\frac{\partial\mathcal{F}[\delta\Phi]}{\partial t} = M\left(\frac{-|\mathbf{k}|^2 - J_f(\Phi^*)}{1-\alpha_0+\alpha_1|\mathbf{k}|^2}\right)\mathcal{F}[\delta\Phi].$$

For stability, the growth rate, which is the real part of the eigenvalues of the matrix in the Fourier domain, must be negative. Therefore, stability is ensured if:

$f(\cdot)$ is increasing near Φ^*, implying that $J_f(\Phi^*)$ has positive eigenvalues, and also $\alpha_1 > 0$, $\alpha_0 < 1$.

This setup assumes that Φ and $f(\Phi)$ can be appropriately generalized to vector fields in multi-dimensional space. The stability analysis involves ensuring that all eigenvalues of the operator in the Fourier-transformed PDE have negative real parts.

2.3 Numerical stability of the system

In this section, we explain the rationale behind adding the perturbation terms and how it is beneficial for the numerical stability.
First consider the unperturbed system

$$\frac{\partial\Phi}{\partial t} = -M(\mathcal{L}\Phi + f(\Phi)).$$

The RHS is generally non-linear. So in order to perform Von-Neumann stability analysis, we need to first linearize it.

Let Φ^* be the steady state solution.

$$\frac{\partial\Phi}{\partial t} = -M\left((\mathcal{L}\Phi^* + f(\Phi^*)) + (\Phi-\Phi^*)\frac{\delta}{\delta\Phi}(\mathcal{L}\Phi + f(\Phi))\Big|_{\Phi^*}\right).$$

$$\frac{\partial\Phi}{\partial t} = -M\left((\Phi-\Phi^*)\lambda^*\right).$$

If we use a forward Euler method, we get:

$$\frac{\Phi^{n+1} - \Phi^n}{\Delta t} = -M\Phi^n\lambda^* + M\Phi^*\lambda^*$$

$$\Phi^{n+1} = (1 - \Delta t M\lambda^*)\,\Phi^n + M\Phi^*\lambda^*$$

For simplicity, we can assume $\Phi^* = 0$ as Φ can be translated to get a zero steady-state solution.

Analyzing the growth factor of the equation, we want: $|1 - \Delta t M\lambda^*| < 1$

$$\implies \Delta t < \frac{2}{M\lambda^*} \tag{2.4}$$

Next consider the type-1 perturbed system:

$$(1 - \alpha_0)\frac{\partial \Phi}{\partial t} = -M(-\nabla^2\Phi + f(\Phi)).$$

From the analysis of physical stability, we need $\alpha_0 < 0$. So letting $\alpha_0 = -\alpha^2$ and linearizing the RHS:

$$(1 + \alpha^2)\frac{\partial \Phi}{\partial t} = -M\left((\Phi - \Phi^*)\lambda^*\right),$$

where $\lambda^* = \frac{\delta}{\delta\Phi}(\mathcal{L}\Phi + f(\Phi))\Big|_{\Phi^*}$.

Using forward Euler and analyzing the growth factor, we need $\left|1 - \Delta t\frac{M\lambda^*}{(1+\alpha^2)}\right| < 1$

$$\implies \Delta t < \frac{2(1+\alpha^2)}{M\lambda^*} \tag{2.5}$$

We can notice that compared to the upper-limit for Δt in Eq. (2.4), we have a higher upper bound for stable numerical scheme in Eq. (2.5). This intuitively explains how perturbing the system leads to a better numerical solvability.

Next consider the type-2 perturbed system:

$$(1 - \alpha_0 - \alpha_1\nabla^2)\frac{\partial \Phi}{\partial t} = -M(\mathcal{L}\Phi + f(\Phi)).$$

Considering in 1d:

$$(1 - \alpha_0)\Phi_t - \alpha_1\Phi_{txx} = -M(\mathcal{L}\Phi + f(\Phi)).$$

Linearizing the RHS around Φ_0 we get:

$$-M(\mathcal{L}\Phi + f(\Phi)) = -M\left((\mathcal{L}\Phi^* + f(\Phi^*)) + (\Phi - \Phi^*)\frac{\delta}{\delta\Phi}(\mathcal{L}\Phi + f(\Phi))\Big|_{\Phi^*}\right)$$

Then discretizing using forward Euler:

$$(1-\alpha_0)\left(\frac{\Phi^{n+1}-\Phi^n}{\Delta t}\right)-\alpha_1\left(\frac{\Phi_{xx}^{n+1}-\Phi_{xx}^n}{\Delta t}\right)=-M\left(\Phi^n-\Phi^*\right)\lambda^*.$$

Since we have spatial derivatives, take the Fourier transform: $\Phi^n=\hat{\Phi}^n e^{ikx}$

$$\hat{\Phi}^{n+1}=\left[1-\frac{M\Delta t\lambda^*}{(1-\alpha_0)+\alpha_1 k^2}\right]\hat{\Phi}^n+M\left(\frac{\phi^*\lambda^*}{(1-\alpha_0)+\alpha_1 k^2}\right)$$

Since we want stablity, the amplitude growth factor must be curbed to less than 1.

$$\left|1-\frac{M\Delta t\lambda^*}{(1-\alpha_0)+\alpha_1 k^2}\right|<1$$

$$\implies \Delta t<\frac{2[(1-\alpha_0)+\alpha_1 k^2]}{M\lambda^*} \tag{2.6}$$

$\therefore$ the upper limits of the time step:

- Unperturbed system:

$$\Delta t<\frac{2}{M\lambda^*}$$

- Type-1 perturbation:

$$\Delta t<\frac{2(1-\alpha_0)}{M\lambda^*}$$

- Type-2 perturbation:

$$\Delta t<\frac{2[(1-\alpha_0)+\alpha_1 k^2]}{M\lambda^*}$$

From the analysis of physical stability we need ti impose $\alpha_0<1,\alpha_1>0$. Thus it can be clearly seen that the adding the perturbation term in the friction operator results in a system with an increased critical time step (CFL limit), thus signaling that it is effective. Also the CFL limit seems to get better by adding increased orders of the the perturbation terms.

2.4 Reformulating Network PDE as a TCPDE system

Consider a bounded domain Ω with piecewise smooth boundary $\partial\Omega$. We consider a PDE system modeling formulation of biological transportation network:

$$\begin{cases} -\nabla \cdot \Big((r\mathbf{I} + \mathbf{n} \otimes \mathbf{n})\nabla p\Big) = S, \\ \mathbf{n}_t - D^2 \Delta \mathbf{n} - c^2(\mathbf{n} \cdot \nabla p)\nabla p + \alpha |\mathbf{n}|^{2(\gamma-1)}\mathbf{n} = \mathbf{0}, \end{cases} \tag{2.7}$$

with:

- Boundary condition: $\mathbf{n} = \mathbf{0}, p = 0$ on $\{x \in \partial\Omega, t > 0\}$.
- Initial condition: $\mathbf{n} = \mathbf{n}_0$ on $\{x \in \Omega, t = 0\}$

$$F[\mathbf{n}, p] = \frac{1}{2}\int_\Omega \left(D^2|\nabla \mathbf{n}|^2 + \frac{|\mathbf{n}|^{2\gamma}}{\gamma} + c^2|\mathbf{n} \cdot \nabla p|^2 + c^2 r|\nabla p|^2 \right) dx, \tag{2.8}$$

We can see that the above energy dissipation functional is negative under these 2 cases of the BCs:

1. $p = p_0(\mathbf{x}), \quad \mathbf{n} = \mathbf{n}_0(\mathbf{x})$
2. $\hat{\mathbf{e}} \cdot \nabla \mathbf{n}_t = 0, \quad p = p_0(\mathbf{x})$

We choose the 1st set of BCs. Upon employing those BCs, the dissipation functional in Eq. (3.4) becomes

$$\begin{aligned} \therefore \tfrac{d}{dt}F(\mathbf{n}(t)) &= -\textstyle\int_\Omega \mathbf{n}_t^2 dx \\ &\leq 0 \end{aligned}$$

Let us reformulate the dissipation functional (2.8).
Multiplying Eq. (2.7) with p and integrating on Ω yields:

$$0 = \int_\Omega \left(|\nabla p|^2 + |\mathbf{n} \cdot \nabla p|^2 - pS \right) dx \tag{2.9}$$

Now, doing (2.8) - $c^2\times$ (2.9), we get:

$$F(p, \mathbf{n}) = \frac{1}{2}\int_\Omega \left(D^2|\nabla \mathbf{n}|^2 + \frac{|\mathbf{n}|^{2\gamma}}{\gamma} - c^2|\mathbf{n} \cdot \nabla p|^2 - c^2 r|\nabla p|^2 + 2c^2 pS \right) d\mathbf{x} \tag{2.10}$$

First we'll consider the 1D spatial domain. Introduce the following notation:

$$\Phi = \begin{bmatrix} p \\ n \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathcal{L} = \begin{bmatrix} 0 & 0 \\ 0 & -D^2\partial_{xx} \end{bmatrix}, \quad \frac{\delta f}{\delta \Phi} = \begin{bmatrix} c^2\partial_x\left[(n^2 + r)p_x\right] + c^2 S \\ n^{2\gamma-1} - c^2 n p_x^2 \end{bmatrix}$$

We can write the network PDE system as this TCPDE system:

$$\begin{cases} C\Phi_t = -M\left[\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi}\right], \\ \frac{d}{dt}F[\phi] = -(\mu, M\mu) = -\left\|n_t^2\right\| \\ F[\phi] = \frac{1}{2}(\phi, \mathcal{L}\phi) + (f(\phi, \nabla\phi), 1) \\ \mu = \mathcal{L}\phi + \frac{\delta f}{\delta \phi} \end{cases} \tag{2.11}$$

We perturb the evolution equation as:

$$PC\Phi_t = -M\left[Q\mathcal{L}\Phi + \frac{\delta f}{\delta \Phi}\right],$$

where we choose $P = Q = (1-\alpha^2)$.

Thus the predictor-corrector scheme:

Scheme 2.1 (NWPDE)**.** Given Φ^{n-1} and Φ^n (and hence $\overline{\Phi}^{n+\frac{1}{2}}$), we compute Φ^{n+1} through the following two steps:

1. Prediction: predict $\Phi_*^{n+\frac{1}{2}}$ via

$$C\left(\frac{\Phi_*^{n+\frac{1}{2}} - \Phi^n}{\tau/2}\right) = -\mathbf{M}\left(\mathcal{L}\Phi_*^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\overline{\Phi}^{n+\frac{1}{2}}\right]\right) \tag{2.12}$$

This is the strategy in this predictor step:

(a) Solve for $n_*^{n+\frac{1}{2}}$ from:

$$\frac{n_*^{n+\frac{1}{2}} - n^n}{\tau/2} = D^2\partial_{xx}n_*^{n+\frac{1}{2}} - \left(\bar{n}^{n+\frac{1}{2}}\right)^{2\gamma-1} + c^2\bar{n}^{n+\frac{1}{2}}\left(\left(\bar{p}^{n+\frac{1}{2}}\right)_x\right)^2$$

This can be solved using a linear solver.

(b) Solve for $p_*^{n+\frac{1}{2}}$ from:

$$c^2\partial_x\left[\left((n_*^{n+\frac{1}{2}})^2 + r\right)(p_*^{n+\frac{1}{2}})_x\right] + c^2 S = 0$$

2. SVM correction:

$$\begin{cases} C(1-\alpha^2)(\delta_t^+\Phi^n) = -\mathbf{M}\left((1-\alpha^2)\mathcal{L}\Phi^{n+\frac{1}{2}} + \frac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right]\right), \\ \delta_t^+ F\left[\Phi^n\right] = -\left(\mu_*^{n+\frac{1}{2}}, \mathbf{M}\mu_*^{n+\frac{1}{2}}\right), \\ F[\Phi] = \dfrac{1}{2}(\Phi, \mathcal{L}\Phi) + (f(\Phi, \nabla\Phi), 1), \\ \mu_*^{n+\frac{1}{2}} = \mathcal{L}\Phi_*^{n+\frac{1}{2}} + \dfrac{\delta f}{\delta \Phi}\left[\Phi_*^{n+\frac{1}{2}}\right] \end{cases} \tag{2.13}$$

Strategy for the corrector step:

(a) Solve for n^{n+1}:

$$\frac{(1-\alpha^2)\big(n^{n+1}-n^n\big)}{\tau} = (1-\alpha^2)\, D^2\, \partial_{xx}\left(\frac{n^{n+1}+n^n}{2}\right) - \left[\big(n_*^{\,n+\frac{1}{2}}\big)^{2\gamma-1} - c^2\, n_*^{\,n+\frac{1}{2}}\big((p_*^{\,n+\frac{1}{2}})_x\big)^2\right]$$

$$\implies \left(1-\tfrac{\tau D^2}{2}\,\partial_{xx}\right) n^{n+1} = \left(1+\tfrac{\tau D^2}{2}\,\partial_{xx}\right) n^n - \frac{\tau}{1-\alpha^2}\left[\big(n_*^{\,n+\frac{1}{2}}\big)^{2\gamma-1} - c^2\, n_*^{\,n+\frac{1}{2}}\big((p_*^{\,n+\frac{1}{2}})_x\big)^2\right].$$

This can be solved as such:

$$\begin{cases} \widehat{n}^{\,n+1} = \left(1-\dfrac{\tau D^2}{2}\partial_{xx}\right)^{-1}\left[\left(1+\dfrac{\tau D^2}{2}\partial_{xx}\right) n^n - \tau\Big((n_*^{\,n+\frac{1}{2}})^{2\gamma-1} - c^2\, n_*^{\,n+\frac{1}{2}}\big((p_*^{\,n+\frac{1}{2}})_x\big)^2\Big)\right], \\ \quad w^n = \left(1-\dfrac{\tau D^2}{2}\partial_{xx}\right)^{-1}\left(1+\dfrac{\tau D^2}{2}\partial_{xx}\right) n^n, \\ \quad \beta = -\alpha^2 \end{cases} \tag{2.14a}$$

$$n^{\,n+1} = \frac{\widehat{n}^{\,n+1} + \beta\, w^n}{1+\beta}.$$

Solve for β using:

$$F[n^{n+1},\, p_*^{\,n+\frac{1}{2}}] = F[n^n,\, p_*^{\,n+\frac{1}{2}}] - \tau \int_\Omega \Big(\mu_{n,*}^{\,n+\frac{1}{2}}\Big)^2\, dx$$

Finally solve for n^{n+1}

(b) Solve for p^{n+1} from:

$$c^2\partial_x\left[\big((n^{n+1})^2 + r\big)\,\frac{\partial}{\partial x}(p^{n+1})\right] + c^2 S = 0,$$

2.5 Alternate: Network PDE using convex splitting

Consider the modified energy functional of the NWPDE system:

$$\mathcal{E}[n,p] = \frac{1}{2}\int_\Omega \left(D^2|\nabla n|^2 + \frac{|n|^{2\gamma}}{\gamma} - c^2(n\cdot\nabla p)^2 - c^2|\nabla p|^2 + 2c^2 pS\right)\, dx. \tag{2.15}$$

For the NWPDE system, this energy is dissipative with the following dissipation property:

$$\frac{d\mathcal{E}}{dt} = -\left\|\frac{\partial n}{\partial t}\right\|^2 = -(\mu_n, \mu_n) \le 0.$$

Now, this energy can be written as the difference of two convex functionals:

$$\mathcal{E}[n,p] = F_1(n,p) - F_2(n,p), \tag{2.16}$$

where

$$F_1(n,p) := \frac{1}{2}\int_\Omega \left(D^2|\nabla n|^2 + \frac{|n|^{2\gamma}}{\gamma} + 2c^2 pS \right) dx, \tag{2.17}$$

$$F_2(n,p) := \frac{1}{2}\int_\Omega \left(c^2(n\cdot\nabla p)^2 + c^2|\nabla p|^2 \right) dx. \tag{2.18}$$

Needed partial derivatives:

$$\frac{\delta F_1}{\delta n} = -D^2\,\Delta n + |n|^{2\gamma-2}\,n, \qquad \frac{\delta F_1}{\delta p} = c^2\,S$$

$$\frac{\delta F_2}{\delta n} = c^2\,(n\cdot\nabla p)\,\nabla p, \qquad \frac{\delta F_2}{\delta p} = -c^2\,\nabla\cdot\big[(n\otimes n + I)\,\nabla p\big]$$

We define the the following gradient flow TCPDE system:

$$\begin{cases} \partial_t n = -\mu_n = -\dfrac{\delta F_1}{\delta n} + \dfrac{\delta F_2}{\delta n} = D^2\,\Delta n - |n|^{2\gamma-2}\,n + c^2\,(n\cdot\nabla p)\,\nabla p, \\ -\alpha^2\partial_t p = -\mu_p = -\dfrac{\delta F_1}{\delta p} + \dfrac{\delta F_2}{\delta p} = -c^2\,S - c^2\,\nabla\cdot\big[(n\otimes n + I)\,\nabla p\big], \\ \mathcal{E}[n,p] = \dfrac{1}{2}\displaystyle\int_\Omega \Big(D^2\,|\nabla n|^2 + \frac{|n|^{2\gamma}}{\gamma} - c^2\,(n\cdot\nabla p)^2 - c^2\,|\nabla p|^2 + 2c^2\,p\,S \Big)\,dx, \\ \dfrac{d\mathcal{E}}{dt} = -(\mu_n,\mu_n) - \alpha^2\,(\mu_p,\mu_p). \end{cases}$$

Note that we introduced the α term on partial derivative of p in the elliptic Darcy-equation, generalizing the systems into a gradient flow. The original NWPDE system is recovered when $\alpha \to 0$.

Scheme 2.2. Given (n^{n-1}, p^{n-1}) and (n^n, p^n), compute n^{n+1} and the half-step pressure $p_*^{n+\frac{1}{2}}$:

1. **Predictor step**:

   Solve for n^{n+1} via:

$$\begin{aligned} D_t^{(2)} n^{n+1} = &-\left(-D^2\Delta n^{n+1} + |n^{n+1}|^{\beta-2} n^{n+1}\right) \\ &+\, c^2\left(2(n^n\cdot\nabla p^n)\,\nabla p^n - (n^{n-1}\cdot\nabla p^{n-1})\,\nabla p^{n-1}\right), \end{aligned} \tag{2.19}$$

   Note that for the n-equation, the convex part is implicit and the concave part is explicit like a typical convex-splitting scheme. However, the concave part uses an Adams Bashforth extrapolation.

Solve for $\Phi_*^{n+\frac{1}{2}}$ via:

$$-c^2\,\nabla\cdot\left[(n^{n+1}\otimes n^{n+1})\nabla p_*^{n+\frac{1}{2}}\right]-c^2S=2\left(-c^2\,\nabla\cdot\left[(n^n\otimes n^n)\nabla p^n\right]-c^2S\right)$$
$$-\left(-c^2\,\nabla\cdot\left[(n^{n-1}\otimes n^{n-1})\nabla p^{n-1}\right]-c^2S\right). \quad (2.20)$$

Here also we use an Adam Bashforth type extrapolation term for second-order $\mathcal{O}(\tau^2)$.

2. **Corrector step**:

$$\mu_{p,*}^{n+\frac{1}{2}}=\frac{\delta F_2}{\delta p}(n^{n+1},p_*^{n+\frac{1}{2}})=-c^2\nabla\cdot\left[(n^{n+1}\otimes n^{n+1})\nabla p_*^{n+\frac{1}{2}}\right]-c^2S$$

Solve for p^{n+1} from:

$$-\alpha^2\left(\frac{3p^{n+1}-4p^n+p^{n-1}}{2\tau}\right)=2\mu_{p,*}^{n+\frac{1}{2}}-\mu_{p,*}^{n-\frac{1}{2}}. \quad (2.21)$$

Strategy to solve p^{n+1}: Rearranging (2.21) to get:

$$p^{n+1}=\frac{4p^n-p^{n-1}}{3}-\frac{2\tau}{3\alpha^2}\left(2\mu_*^{n+\frac{1}{2}}-\mu_*^{n-\frac{1}{2}}\right).$$

Letting:

$$\beta=\tau\alpha^2, \quad (2.22)$$
$$\widehat{p}^{n+1}=\frac{4p^n-p^{n-1}}{3}, \quad (2.23)$$
$$w^n=-\frac{2\tau^2}{3}\left(2\mu_*^{n+\frac{1}{2}}-\mu_*^{n-\frac{1}{2}}\right). \quad (2.24)$$

Note that here $\widehat{p}^{n+1}$ and w^n can be directly computed without the need for any linear solvers.

We get:

$$p^{n+1}(\beta)=\widehat{p}^{n+1}+\frac{1}{\beta}w^n.$$

Using the dissipation equation:

$$F\left[\widehat{p}^{n+1}+\beta^{-1}w^n\right]=F[p^n]\left[-\tau\left(\mu_{n,*}^{n+\frac{1}{2}},\mu_{n,*}^{n+\frac{1}{2}}\right)-\beta\left(\mu_{n,*}^{p+\frac{1}{2}},\mu_{p,*}^{p+\frac{1}{2}}\right)\right]. \quad (2.25)$$

Solve this one-dimensional nonlinear equation for β.

$$p^{n+1}\leftarrow\widehat{p}^{n+1}+\beta^{-1}w^n,\qquad \alpha^{n+1}\leftarrow\sqrt{\beta/\tau}.$$

Network generating PDE (older notes)

Consider a bounded domain Ω with piecewise smooth boundary $\partial\Omega$. We consider a PDE system modeling formulation of biological transportation network:

$$\begin{cases} -\nabla\cdot\Big((r\mathbf{I}+\mathbf{n}\otimes\mathbf{n})\nabla p\Big)=S, \\ \mathbf{n}_t-D^2\Delta\mathbf{n}-c^2(\mathbf{n}\cdot\nabla p)\nabla p+\alpha|\mathbf{n}|^{2(\gamma-1)}\mathbf{n}=\mathbf{0}, \end{cases} \tag{2.26}$$

with:

- Boundary condition: $\mathbf{n}=\mathbf{0}, p=0$ on $\{x\in\partial\Omega, t>0\}$.
- Initial condition: $\mathbf{n}=\mathbf{n}_0$ on $\{x\in\Omega, t=0\}$

2.5.1 Energy law

We define the energy functional of the above system as:

$$\mathcal{E}[\mathbf{n},p]=\frac{1}{2}\int_\Omega\left(D^2|\nabla\mathbf{n}|^2+\alpha\frac{|\mathbf{n}|^{2\gamma}}{\gamma}+c^2|\mathbf{n}\cdot\nabla p|^2+c^2r|\nabla p|^2\right)dx, \tag{2.27}$$

where D,α,r,γ,c are positive constants. We multiply (2.7) by p and integrate over Ω to obtain

$$-\int_{\partial\Omega}(r\mathbf{I}+\mathbf{nn})\cdot\nabla p\mathbf{e}ds+\int_\Omega[(r\mathbf{I}+\mathbf{nn}):\nabla p\nabla p-Sp]d\mathbf{x}=0. \tag{2.28}$$

With the boundary condition given above, we have

$$\int_\Omega[(r\mathbf{I}+\mathbf{nn}):\nabla p\nabla p-Sp]d\mathbf{x}=0. \tag{2.29}$$

Applying the equality to the energy function, we obtain another form of the energy:

$$\mathcal{E}[\mathbf{n},p]=\frac{1}{2}\int_\Omega\left(D^2|\nabla\mathbf{n}|^2+\alpha\frac{|\mathbf{n}|^{2\gamma}}{\gamma}-c^2|\mathbf{n}\cdot\nabla p|^2-c^2r|\nabla p|^2+2c^2Sp\right)dx \tag{2.30}$$

2.6 Using SVM on the network PDE system

$$\begin{cases} \mathbf{n}_t=D^2\Delta\mathbf{n}+c^2(\mathbf{n}\cdot\nabla p)\nabla p-\alpha|\mathbf{n}|^{2(\gamma-1)}\mathbf{n}+\Theta g_1[\mathbf{n},p], \\ -\nabla\cdot\Big((r\mathbf{I}+\mathbf{n}\otimes\mathbf{n})\nabla p\Big)=S, \\ \mathcal{E}[\mathbf{n},p]=\frac{1}{2}\int_\Omega\left(D^2|\nabla\mathbf{n}|^2+\alpha\frac{|\mathbf{n}|^{2\gamma}}{\gamma}+c^2|\mathbf{n}\cdot\nabla p|^2+c^2r|\nabla p|^2\right)dx+\Theta g_2[\mathbf{n},p], \\ \frac{d}{dt}\mathcal{E}[\mathbf{n}(t,p])=-\int_\Omega\mathbf{n}_t^2=-\int_\Omega\left(D^2\Delta\mathbf{n}+c^2(\mathbf{n}\cdot\nabla p)\nabla p-\alpha|\mathbf{n}|^{2(\gamma-1)}\mathbf{n}+\Theta g_3[\mathbf{n},p]\right)^2dx \end{cases} \tag{2.31}$$

Now we have 4 unknowns $\{\mathbf{n},p,\mathcal{E},\Theta\}$ and 4 equations, which makes it a consistent system.

Alternate approach: We can reformulate the model as a limiting case of a gradient flow.

$$\begin{cases} \mathbf{n}_t = -\mu_{\mathbf{n}} + \Theta g_1[\mathbf{n}, p], \\ \mu_{\mathbf{n}} = -D^2 \Delta \mathbf{n} - c^2 (\mathbf{n} \cdot \nabla p) \nabla p + \alpha |\mathbf{n}|^{2(\gamma-1)} \mathbf{n} \\ \eta p_t = -(c^2 \nabla \cdot \Big((r\mathbf{I} + \mathbf{n} \otimes \mathbf{n}) \nabla p \Big) + c^2 S) + \Theta g_2[\mathbf{n}, p], \\ \mathcal{E}[\mathbf{n}, p] = \frac{1}{2} \int_\Omega \Big(D^2 |\nabla \mathbf{n}|^2 + \alpha \frac{|\mathbf{n}|^{2\gamma}}{\gamma} + c^2 |\mathbf{n} \cdot \nabla p|^2 + c^2 r |\nabla p|^2 \Big) \, dx + \Theta g_3[\mathbf{n}, p], \\ \frac{d}{dt} \mathcal{E}[\mathbf{n}(t, p]) = -\int_\Omega \mathbf{n}_t^2 = -\int_\Omega \Big(D^2 \Delta \mathbf{n} + c^2 (\mathbf{n} \cdot \nabla p) \nabla p - \alpha |\mathbf{n}|^{2(\gamma-1)} \mathbf{n} + \Theta g_4[\mathbf{n}, p] \Big)^2 \, dx \end{cases} \tag{2.32}$$

Here we have 6 unknowns $\{\mathbf{n}, \mu_{\mathbf{n}}, p, \mu_p, \mathcal{E}, \Theta\}$ and 4 equations, which makes it a consistent system.

3 Using convex splitting way

$$\mathcal{E}[\mathbf{n}, p] = \frac{1}{2}\int_\Omega \left(D^2|\nabla \mathbf{n}|^2 + \alpha\frac{|\mathbf{n}|^{2\gamma}}{\gamma} - c^2|\mathbf{n}\cdot\nabla p|^2 - c^2 r|\nabla p|^2 + 2c^2 Sp \right) dx \tag{3.1}$$

In this free energy expression, we define

$$f_1(\mathbf{n}) = \frac{\alpha}{2}\frac{|\mathbf{n}|^{2\gamma}}{\gamma}, \quad f_2(\mathbf{n}, p) = \frac{c^2}{2}[|\mathbf{n}\cdot\nabla p|^2 + r|\nabla p|^2 - 2Sp]. \tag{3.2}$$

We note that both are convex. Thus, the constrained free energy in the form is given by a difference of two convex functions. Whereas we conclude from the first expression that the free energy is convex in both $\mathbf{n}$ and p.

The variation of the free energy with respect to $\mathbf{n}$ and p are given as follows

$$\frac{\delta\mathcal{E}}{\delta\mathbf{n}} = -D\nabla^2\mathbf{n} + \frac{\partial}{\partial\mathbf{n}}(f_1 - f_2), \quad \frac{\delta\mathcal{E}}{\delta p} = 0. \tag{3.3}$$

Finding the energy dissipation rate:

$$\frac{d}{dt}\mathcal{E}[\mathbf{n}(t), p] = -\int_\Omega (\partial_t\mathbf{n})^2 + \int_{\partial\Omega}[D^2\hat{\mathbf{e}}\mathbf{n} : \nabla\mathbf{n}_t - c^2\hat{\mathbf{e}}\cdot(r\mathbf{I} + \mathbf{n}\otimes\mathbf{n})\cdot\nabla p p_t]dS \tag{3.4}$$

where $\hat{\mathbf{e}}$ is the outward unit normal at the boundary $\partial\Omega$.

We can see that the above functional is negative under these 2 cases of the BCs:

1. $p = p_0(\mathbf{x}), \quad \mathbf{n} = \mathbf{n}_0(\mathbf{x})$
2. $\hat{\mathbf{e}}\cdot\nabla\mathbf{n}_t = 0, \quad p = p_0(\mathbf{x})$

We choose the 1st set of BCs. Upon employing those BCs, the dissipation functional in Eq. (3.4) becomes

$$\tfrac{d}{dt}\mathcal{E}[\mathbf{n}(t, p]) \quad = -\textstyle\int_\Omega \mathbf{n}_t^2 dx \leq 0$$

3.1 Convex splitting method for Network generating PDEs

Introducing an L_2 gradient flow over $\mathbf{n}$ for the reformulated energy functional (**??**):

$$\partial_t\mathbf{n} = -\tfrac{\delta\mathcal{E}}{\delta\mathbf{n}}, \quad \tfrac{\delta\mathcal{E}}{\delta\mathbf{n}} = -D^2\nabla^2\mathbf{n} + \mathbf{f}_1'(\mathbf{n}) - \mathbf{f}_2'(\mathbf{n}, p).$$

The PDE system is given by

$$\begin{aligned} &\partial_t\mathbf{n} = -\tfrac{\delta\mathcal{E}}{\delta\mathbf{n}} = D^2\nabla^2\mathbf{n} - \mathbf{f}_1'(\mathbf{n}) + \mathbf{f}_2'(\mathbf{n}, p), \\ &\tfrac{\delta\mathcal{E}}{\delta p} = 0. \end{aligned} \tag{3.5}$$

The second equation serves as a constraint

$$-\nabla \cdot \Big((r\mathbf{I} + \mathbf{n} \otimes \mathbf{n}) \nabla p \Big) - S = 0. \tag{3.6}$$

In fact, this equation can be regarded as the Euler-Lagrange equations for the optimization problem for any given $\mathbf{n}$:

$$\max_p \mathcal{E}[\mathbf{n}, p]. \tag{3.7}$$

The gradient flow equation can be regarded as a minimization problem for the Rayleighian under the above constraint:

$$\min_{\mathbf{n}_t} [\mathcal{E}_t + \frac{1}{2} \int_\Omega \|\mathbf{n}_t\|^2 d\mathbf{x}]. \tag{3.8}$$

Now we propose a discretized scheme:

$$\left.\begin{aligned} \frac{\mathbf{n}^{n+1} - \mathbf{n}^n}{\Delta t} &= D^2 \nabla^2 \mathbf{n}^{n+1} - \mathbf{f}_1'(\mathbf{n}^{n+1}) + \mathbf{f}_2'(\mathbf{n}^n, p^{n+1}) \\ -\nabla \cdot \Big((r\mathbf{I} + \mathbf{n}^n \otimes \mathbf{n}^n) \nabla p^{n+1} \Big) &= S \end{aligned}\right\} \tag{3.9}$$

Note that the Darcy-type equation above is treated as an ancillary equation. In the time-marching method, first we solve that explicitly for p^{n+1} and then later solve for $\mathbf{n}^{n+1}$ implicitly.

From earlier we have:

$$\mathcal{E}[\mathbf{n}, p] = \frac{1}{2} \int_\Omega \left(D^2 |\nabla \mathbf{n}|^2 + f_1(\mathbf{n}) - f_2(\mathbf{n}, p) \right) dx.$$

We replace $\mathbf{f}_2(\mathbf{n}, p)$ by its linear approximation about $\mathbf{n}^n$. Using this we define a modified energy

$$\begin{aligned} \tilde{\mathcal{E}}(\mathbf{n}; \mathbf{n}^n, p^{n+1}) \quad &:= \tfrac{1}{2} \textstyle\int_\Omega \left(D^2 |\nabla \mathbf{n}|^2 + f_1(\mathbf{n}) - (f_2(\mathbf{n}^n, p^{n+1}) + \nabla f_2(\mathbf{n}^n, p^{n+1}) \cdot (\mathbf{n} - \mathbf{n}^n)) \right) dx \\ &+ \frac{1}{2\Delta t} \textstyle\int_\Omega |\mathbf{n} - \mathbf{n}^n|^2 dx \end{aligned}$$

Now we introduce the following energy minimization problem:

$$\mathbf{n}^{n+1} = \arg\min_{\mathbf{n}} \quad \tilde{\mathcal{E}}(\mathbf{n}, \mathbf{n}^n, p^{n+1}) \tag{3.10}$$

and claim that its solution is identical to the numerical scheme (3.9).

Theorem 3.1. *The dynamics defined by the minimization problem (3.10) or equivalently the discretized system (3.9) obey dissipation property*

$$\mathcal{E}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) \leq \mathcal{E}(\mathbf{n}^n; \mathbf{n}^n, p^{n+1})$$

Proof. Consider the discrete energy we introduced earlier at $\mathbf{n} = \mathbf{n}^{n+1}$:

$$\begin{aligned}\tilde{\mathcal{E}}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) &= \tfrac{1}{2}\int_\Omega \left(D^2|\nabla \mathbf{n}^{n+1}|^2 + f_1(\mathbf{n}^{n+1}) - (f_2(\mathbf{n}^n, p) + \nabla f_2(\mathbf{n}^n, p)\cdot(\mathbf{n}^{n+1} - \mathbf{n}^n))\right) dx \\ &+ \frac{1}{2\Delta t}\int_\Omega |\mathbf{n}^{n+1} - \mathbf{n}^n|^2 dx.\end{aligned}$$

Since f_2 is a convex function, its linear approximation (tangent) evaluated at any point is less than or equal to f_2 at that point. Hence $\left(f_2(\mathbf{n}^n, p^{n+1}) + \nabla f_2(\mathbf{n}^n, p^{n+1})\cdot(\mathbf{n} - \mathbf{n}^n)\right) \leq f_2(\mathbf{n}, p^{n+1})$. We let $\mathbf{n} = \mathbf{n}^{n+1}$ and use its converse to get the following.

$$\begin{aligned}\tilde{\mathcal{E}}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) &\geq \tfrac{1}{2}\int_\Omega \left(D^2|\nabla \mathbf{n}^{n+1}|^2 + f_1(\mathbf{n}^{n+1}) - f_2(\mathbf{n}^{n+1}, p^{n+1})\right) dx \\ &+ \frac{1}{2\Delta t}\int_\Omega |\mathbf{n}^{n+1} - \mathbf{n}^n|^2 dx \\ &= \mathcal{E}(\mathbf{n}^{n+1}) + \frac{1}{2\Delta t}\int_\Omega |\mathbf{n}^{n+1} - \mathbf{n}^n|^2 dx\end{aligned}$$

$$\therefore \tilde{\mathcal{E}}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) \geq \mathcal{E}(\mathbf{n}^{n+1}) + \frac{1}{2\Delta t}\int_\Omega |\mathbf{n}^{n+1} - \mathbf{n}^n|^2 dx \tag{3.11}$$

By definition of the problem (3.10), $\mathbf{n}^{n+1}$ is the minimizer of $\tilde{\mathcal{E}}(\mathbf{n}, \mathbf{n}^n)$

$$\therefore \tilde{\mathcal{E}}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) \leq \tilde{\mathcal{E}}(\mathbf{n}; \mathbf{n}^n, p^{n+1}), \forall \mathbf{n}$$

Choosing $\mathbf{n} = \mathbf{n}^n$,

$$\begin{aligned}\tilde{\mathcal{E}}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) &\leq \tilde{\mathcal{E}}(\mathbf{n}^n; \mathbf{n}^n, p^{n+1}) \\ &= \tfrac{1}{2}\int_\Omega \left(D^2|\nabla \mathbf{n}^n|^2 + f_1(\mathbf{n}^n) - f_2(\mathbf{n}^n, p^{n+1})\right) dx \\ &= \mathcal{E}(\mathbf{n}^n, p^{n+1})\end{aligned}$$

$$\therefore \tilde{\mathcal{E}}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) \leq \mathcal{E}(\mathbf{n}^n, p^{n+1}) \tag{3.12}$$

Thus from (3.11) and (3.12), we have:

$$\mathcal{E}(\mathbf{n}^{n+1}, p^{n+1}) + \frac{1}{2\Delta t}\int_\Omega |\mathbf{n}^{n+1} - \mathbf{n}^n|^2 dx \leq \tilde{\mathcal{E}}(\mathbf{n}^{n+1}; \mathbf{n}^n, p^{n+1}) \leq \tilde{\mathcal{E}}(\mathbf{n}^n; \mathbf{n}^n, p^{n+1}) = \mathcal{E}(\mathbf{n}^n, p^{n+1})$$

Hence the discretized system (3.9) respects an energy dissipation law. □

Next we seek to prove true energy dissipation going from n^{th} time-step to the $(n+1)^{th}$.

Theorem 3.2.

$$\mathcal{E}(\mathbf{n}^{n+1}, p^{n+1}) \leq \mathcal{E}(\mathbf{n}^n, p^n)$$

Proof. We know

$$\mathcal{E}(\mathbf{n}^n, p^n) = \frac{D^2}{2}\|\nabla \mathbf{n}^n\|^2 + (f_1(\mathbf{n}^n), 1) - (f_2(\mathbf{n}^n, p^n), 1)$$

Here $(\cdot, \cdot)$ is used to notate inner product over Ω.

For the 1st equation in the discretized system (3.9), multiply with $\Delta t(\mathbf{n}^{n+1}-\mathbf{n}^n)$ and take inner product.

$$\begin{aligned}\left\|\mathbf{n}^{n+1}-\mathbf{n}^n\right\|^2 &= D^2(\nabla^2\mathbf{n}^{n+1},\mathbf{n}^{n+1}) - D^2(\nabla^2\mathbf{n}^{n+1},\mathbf{n}^n) - (f_1'(\mathbf{n}^{n+1}),\mathbf{n}^{n+1}-\mathbf{n}^n) + (\tfrac{\partial f_2}{\partial \mathbf{n}}(\mathbf{n}^n,p^n),\mathbf{n}^{n+1}-\mathbf{n}^n)\\ &= -D^2(\nabla\mathbf{n}^{n+1},\nabla\mathbf{n}^{n+1}) + D^2(\nabla\mathbf{n}^{n+1},\nabla\mathbf{n}^n) - (f_1'(\mathbf{n}^{n+1}),\mathbf{n}^{n+1}-\mathbf{n}^n) + (\tfrac{\partial f_2}{\partial \mathbf{n}}(\mathbf{n}^n,p^n),\mathbf{n}^{n+1}-\mathbf{n}^n)\end{aligned}$$

<u>We use the following identities</u>:

- $(\nabla\mathbf{n}^{n+1},\nabla\mathbf{n}^n) = \frac{1}{2}[\left\|\nabla\mathbf{n}^{n+1}\right\|^2 + \|\nabla\mathbf{n}^n\|^2 - \left\|\nabla\mathbf{n}^{n+1}-\nabla\mathbf{n}^n\right\|^2]$
- Taylor exp of $f_1(\mathbf{n}^n)$ about $\mathbf{n}^{n+1}$:

$$f_1(\mathbf{n}^n) = f_1(\mathbf{n}^{n+1}) + f_1'(\mathbf{n}^{n+1})\cdot(\mathbf{n}^n-\mathbf{n}^{n+1}) + \frac{1}{2}(\mathbf{n}^n-\mathbf{n}^{n+1})^T\cdot f_1''(\xi_1^{n+1})\cdot(\mathbf{n}^n-\mathbf{n}^{n+1})$$

$$\implies (f_1'(\mathbf{n}^{n+1}),\mathbf{n}^{n+1}-\mathbf{n}^n) = -(f_1(\mathbf{n}^n),1)+(f_1(\mathbf{n}^{n+1}),1)+\frac{1}{2}\left((\mathbf{n}^n-\mathbf{n}^{n+1})^T\cdot f_1''(\xi_1^{n+1}),\mathbf{n}^n-\mathbf{n}^{n+1}\right)$$

- Taylor exp of $f_2(\mathbf{n}^{n+1},p^{n+1})$ about $\mathbf{n}^n$:

$$f_2(\mathbf{n}^{n+1},p^{n+1}) = f_2(\mathbf{n}^n,p^{n+1})+\frac{\partial f_2}{\partial \mathbf{n}}(\mathbf{n}^n,p^{n+1})\cdot(\mathbf{n}^{n+1}-\mathbf{n}^n)+\frac{1}{2}(\mathbf{n}^{n+1}-\mathbf{n}^n)^T\cdot\frac{\partial^2 f_2}{\partial \mathbf{n}^2}(\xi_2^n,p^{n+1})\cdot(\mathbf{n}^{n+1}-\mathbf{n}^n)$$

$$\begin{aligned}\implies (\tfrac{\partial f_2}{\partial \mathbf{n}}(\mathbf{n}^n,p^{n+1}),\mathbf{n}^{n+1}-\mathbf{n}^n) = \quad & (f_2(\mathbf{n}^{n+1},p^{n+1}),1) - (f_2(\mathbf{n}^n,p^{n+1}),1)\\ & -\tfrac{1}{2}\left((\mathbf{n}^{n+1}-\mathbf{n}^n)^T\cdot\tfrac{\partial^2 f_2}{\partial \mathbf{n}^2}(\xi_2^n,p^{n+1}),\mathbf{n}^{n+1}-\mathbf{n}^n\right)\end{aligned}$$

□

$$\begin{aligned}\left\|\mathbf{n}^{n+1}-\mathbf{n}^n\right\|^2 &= -D^2\left\|\nabla\mathbf{n}^{n+1}\right\|^2 + \tfrac{D^2}{2}\left[\left\|\nabla\mathbf{n}^{n+1}\right\|^2 + \|\nabla\mathbf{n}^n\|^2 - \left\|\nabla\mathbf{n}^{n+1}-\nabla\mathbf{n}^n\right\|^2\right]\\ &\quad - \left[-(f_1(\mathbf{n}^n),1) + f_1(\mathbf{n}^{n+1},1) + \tfrac{1}{2}\left((\mathbf{n}^n-\mathbf{n}^{n+1})^T\cdot f_1''(\xi_1^{n+1}),\mathbf{n}^n-\mathbf{n}^{n+1}\right)\right]\\ &\quad + \left[(f_2(\mathbf{n}^{n+1},p^{n+1}),1) - (f_2(\mathbf{n}^n,p^{n+1}),1) - \tfrac{1}{2}\left((\mathbf{n}^{n+1}-\mathbf{n}^n)^T\cdot\tfrac{\partial^2 f_2}{\partial \mathbf{n}^2}(\xi_2^n,p^{n+1}),\mathbf{n}^{n+1}-\mathbf{n}^n\right)\right]\end{aligned}$$

Rearranging the terms:

$$\begin{aligned}\left[\tfrac{D^2}{2}\left\|\nabla\mathbf{n}^{n+1}\right\|^2 + f_1(\mathbf{n}^{n+1},1) - (f_2(\mathbf{n}^{n+1},p^{n+1}),1)\right] = \quad & -\tfrac{D^2}{2}\left\|\mathbf{n}^{n+1}-\mathbf{n}^n\right\|^2\\ -\left[\tfrac{D^2}{2}\|\nabla\mathbf{n}^n\|^2 + (f_1(\mathbf{n}^n),1) - (f_2(\mathbf{n}^n,p^{n+1}),1)\right] \quad & -\tfrac{1}{2}\left((\mathbf{n}^n-\mathbf{n}^{n+1})^T\cdot f_1''(\xi_1^{n+1}),\mathbf{n}^n-\mathbf{n}^{n+1}\right)\\ & -\tfrac{1}{2}\left((\mathbf{n}^{n+1}-\mathbf{n}^n)^T\cdot\tfrac{\partial^2 f_2}{\partial \mathbf{n}^2}(\xi_2^n,p^{n+1}),\mathbf{n}^{n+1}-\mathbf{n}^n\right)\end{aligned}$$

In the 2nd energy term, instead of $(f_2(\mathbf{n}^n,p^{n+1}),1)$ we need $(f_2(\mathbf{n}^n,p^n),1)$.

Using Taylor exp of $f_2(\mathbf{n}^n, p^{n+1})$ about p^n:

$$(f_2(\mathbf{n}^n, p^{n+1}), 1) = (f_2(\mathbf{n}^n, p^n), 1) + (\frac{\partial f_2}{\partial p}(\mathbf{n}^n, p^n), p^{n+1} - p^n) + \frac{1}{2}(\frac{\partial^2 f_2}{\partial p^2}(\mathbf{n}^n, \xi_3^n)(p^{n+1} - p^n), (p^{n+1} - p^n))$$

$$\begin{aligned}
&\left[\tfrac{D^2}{2}\left\|\nabla \mathbf{n}^{n+1}\right\|^2 + f_1(\mathbf{n}^{n+1}, 1) - (f_2(\mathbf{n}^{n+1}, p^{n+1}), 1)\right] = && -\tfrac{D^2}{2}\left\|\mathbf{n}^{n+1} - \mathbf{n}^n\right\|^2 \\
&- \left[\tfrac{D^2}{2}\left\|\nabla \mathbf{n}^n\right\|^2 + (f_1(\mathbf{n}^n), 1) - (f_2(\mathbf{n}^n, p^n), 1)\right] && -\tfrac{1}{2}\left((\mathbf{n}^{n+1} - \mathbf{n}^n)^T \cdot f_1''(\xi_1^{n+1}), \mathbf{n}^{n+1} - \mathbf{n}^n\right) \\
&+ (\tfrac{\partial f_2}{\partial p}(\mathbf{n}^n, p^n), p^{n+1} - p^n) && -\tfrac{1}{2}\left((\mathbf{n}^{n+1} - \mathbf{n}^n)^T \cdot \tfrac{\partial^2 f_2}{\partial \mathbf{n}^2}(\xi_2^n, p^{n+1}), \mathbf{n}^{n+1} - \mathbf{n}^n\right) \\
& && -\tfrac{1}{2}(\tfrac{\partial^2 f_2}{\partial p^2}(\mathbf{n}^n, \xi_3^n)(p^{n+1} - p^n), (p^{n+1} - p^n))
\end{aligned}$$

Since f_1 and f_2 are convex w.r.t their arguments, their second derivatives are positive or their Hessians are positive definite. Hence all the 3 error terms of Taylor exp on the RHS are non-positive.

We have:

$$\mathcal{E}(\mathbf{n}^{n+1}, p^{n+1}) - \mathcal{E}(\mathbf{n}^n, p^n) + (\tfrac{\partial f_2}{\partial p}(\mathbf{n}^n, p^n), p^{n+1} - p^n) \leq \quad 0$$

<u>Need to investigate the sign of $(\frac{\partial f_2}{\partial p}(\mathbf{n}^n, p^n), p^{n+1} - p^n)$:</u>

$$\begin{aligned}
(\tfrac{\partial f_2}{\partial p}(\mathbf{n}^n, p^n), p^{n+1} - p^n) &= c^2(-\nabla \cdot \Big((r\mathbf{I} + \mathbf{n}^n \otimes \mathbf{n}^n)\nabla p^n\Big) - S, p^{n+1} - p^n) \\
&= c^2(-r\nabla \cdot (\nabla p^n), p^{n+1} - p^n) - c^2(\nabla \cdot (\mathbf{n} \otimes \mathbf{n}\nabla p^n), p^{n+1} - p^n) - c^2(S, p^{n+1} - p^n) \\
&= c^2(r\nabla p^n, \nabla p^{n+1} - \nabla p^n) + c^2(\mathbf{n} \otimes \mathbf{n}\nabla p^n, \nabla p^{n+1} - \nabla p^n) - c^2(S, p^{n+1} - p^n)
\end{aligned} \tag{3.13}$$

Consider the 2nd equation from our discretized system (3.9)

$$-\nabla \cdot \Big((r\mathbf{I} + \mathbf{n}^n \otimes \mathbf{n}^n)\nabla p^{n+1}\Big) - S = 0$$

Multiply by $c^2(p^{n+1} - p^n)$ and find the inner product:

$$\begin{aligned}
0 = \quad & c^2(-\nabla \cdot \Big((r\mathbf{I} + \mathbf{n}^n \otimes \mathbf{n}^n)\nabla p^{n+1}\Big) - S, p^{n+1} - p^n) \\
&= c^2(-r\nabla \cdot (\nabla p^{n+1}), p^{n+1} - p^n) - c^2(\nabla \cdot (\mathbf{n} \otimes \mathbf{n}\nabla p^{n+1}), p^{n+1} - p^n) - c^2(S, p^{n+1} - p^n) \\
&= c^2(r\nabla p^{n+1}, \nabla p^{n+1} - \nabla p^n) + c^2(\mathbf{n} \otimes \mathbf{n}\nabla p^{n+1}, \nabla p^{n+1} - \nabla p^n) - c^2(S, p^{n+1} - p^n)
\end{aligned} \tag{3.14}$$

Subtracting (3.14) from (3.13). (3.13) -(3.14) $\implies$:

$$\begin{aligned}
(\tfrac{\partial f_2}{\partial p}(\mathbf{n}^n, p^n), p^{n+1} - p^n) &= c^2 r(\nabla p^n - \nabla p^{n+1}, \nabla p^{n+1} - \nabla p^n) + c^2((\mathbf{n} \otimes \mathbf{n})(\nabla p^n - \nabla p^{n+1}), \nabla p^{n+1} - \nabla p^n) \\
&= -c^2 r\left\|\nabla p^{n+1} - \nabla p^n\right\|^2 - c^2\left\|\mathbf{n}^n \cdot (\nabla p^{n+1} - \nabla p^n)\right\|^2
\end{aligned} \tag{3.15}$$

$$
\begin{aligned}
\mathcal{E}(\mathbf{n}^{n+1}, p^{n+1}) - \mathcal{E}(\mathbf{n}^n, p^n) = \quad & -\tfrac{D^2}{2} \left\| \mathbf{n}^{n+1} - \mathbf{n}^n \right\|^2 \\
& -\tfrac{1}{2} \left((\mathbf{n}^{n+1} - \mathbf{n}^n)^T \cdot f_1''(\xi_1^{n+1}), \mathbf{n}^{n+1} - \mathbf{n}^n \right) \\
& -\tfrac{1}{2} \left((\mathbf{n}^{n+1} - \mathbf{n}^n)^T \cdot \tfrac{\partial^2 f_2}{\partial \mathbf{n}^2}(\xi_2^n, p^{n+1}), \mathbf{n}^{n+1} - \mathbf{n}^n \right) \\
& -\tfrac{1}{2} (\tfrac{\partial^2 f_2}{\partial p^2}(\mathbf{n}^n, \xi_3^n)(p^{n+1} - p^n), (p^{n+1} - p^n)) \\
& +c^2 r \left\| \nabla p^{n+1} - \nabla p^n \right\|^2 + c^2 \left\| \mathbf{n}^n \cdot (\nabla p^{n+1} - \nabla p^n) \right\|^2
\end{aligned}
$$

In the next section we proceed to solve our discrete scheme (3.9) numerically.

4 Numerical simulations

commented out the figures, etc.